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FINAL DEGREE PROJECT

Bioconversion of vegetable food waste: Enzymatic liquefaction and acetic fermentation for vinegar development (Case study: Geranium)

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Gastronomy and Culinary Arts Degree

FINAL DEGREE PROJECT

BIOCONVERSION OF VEGETABLE FOOD WASTE: ENZYMATIC LIQUEFACTION AND ACETIC FERMENTATION FOR VINEGAR DEVELOPMENT (CASE STUDY: GERANIUM)

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COMPANY WHERE THE PROJECT TOOK PLACE: Geranium

DEFENSE DATE: Donostia, on thirty of June 2026.

DECLARATION OF ORIGINALTY

I David Revoltós Rosique

I declare that this Final Degree Project is original, the result of my personal work and that it has not been previously submitted to obtain another degree or professional qualification.

The ideas, formulations, images, illustrations taken from outside sources have been duly cited and referenced.

RESUMEN

Este proyecto explora el desarrollo e implementación de un método de licuefacción enzimática vegetal para unir la gestión de residuos industriales y la alta cocina. Esta técnica procesa sustratos crudos y cocidos, incluyendo recortes.

La investigación detalla la optimización de los parámetros del proceso para garantizar consistencia de sabor y reproducibilidad en diversas familias botánicas. Aplicando fermentación acética a los líquidos resultantes, el estudio demuestra la creación de productos culinarios de alto valor, como vinagres artesanales, alineados con la narrativa y objetivos económicos de una cocina profesional. Esta investigación establece un marco de gastronomía circular, dando a los chefs una herramienta versátil para minimizar el desperdicio alimentario y potenciar las posibilidades creativas mediante la ciencia de los alimentos. Este documento describe la metodología, viabilidad y potencial de la degradación enzimática como técnica culinaria transformadora de residuos alimentarios.

Palabras clave*: gastronomía circular; enzimas; bioprocesamiento; plant-based; fermentación acética.*

- ODS 2: Hambre cero; aprovechamiento de mermas y subproductos.
- ODS 9: Industria, innovación e infraestructura; desarrollo de técnicas culinarias para el tratamiento de residuos.
- ODS 12: Producción y consumo responsables; promoción del uso integral de los productos.

LABURPENA

Proiektu honek barazkien entzima-likefazio metodo baten garapena eta ezarpena aztertzen ditu, hondakin industrialen kudeaketaren eta goi-mailako gastronomiaren arteko zubia egiteko diseinatua. Teknika honek substratu gordinak zein prestatuak prozesatzea ahalbidetzen du, moztutako hondarrak barne.

Ikerketak prozesuko parametroen optimizazioa xehatzen du, zaporearen konstantzia eta familia botaniko desberdinetan erreproduzigarritasuna bermatzeko. Lortutako likidoei hartzidura azetikoa aplikatuz, ikerketak balio handiko produktu kulinarioen sorrera erakusten du, hala nola binagre artisauak, sukalde profesional baten kontakizunarekin (*storytelling*) eta helburu ekonomikoekin bat datozenak. Azkenik, ikerketa honek gastronomia zirkularrerako esparru bat ezartzen du, sukaldariei elikagaien hondakinak minimizatzeko tresna polifazetikoa eskainiz eta, aldi berean, elikagaien zientziaren bidez sormen-aukerak areagotuz. Dokumentu honek degradazio entzimatikoaren metodologia, bideragarritasuna eta teknika kulinario eraldatzaile gisa duen potentziala azaltzen ditu.

Gako-hitzak: gastronomia zirkularra; entzimak; bioprozesamendua; landare-jatorrikoak; hartzidura azetikoa.

- 2. GJHe: Zero gosea; ebakinen eta azpiproduktuen aprobetxamendua.
- 9. GJHe: Industria, berrikuntza eta azpiegitura; hondakinak tratatzeko teknika kulinarioaren garapena.
- 12. GJHe: Ekoizpen eta kontsumo arduratsuak; produktuen erabilera integrala sustatzea.

SUMMARY

This project explores the development and implementation of an enzymatic liquefaction method for vegetables, designed to bridge the gap between industrial waste management and fine dining. This technique allows for the processing of both raw and cooked substrates, including trimmings.

The research details the optimization of the process parameters to ensure flavor consistency and reproducibility across various botanical families. By applying acetic fermentation to the resulting liquids, the study demonstrates the creation of high-value culinary products, such as artisanal vinegars, that align with the storytelling and economic goals of a professional kitchen. Ultimately, this investigation establishes a framework for circular gastronomy, providing chefs with a versatile tool to minimize food waste while enhancing creative possibilities through food science. This document outlines the methodology, feasibility, and the potential of enzymatic degradation as a transformative; food waste; culinary technique.

Keywords: *circular gastronomy*; *enzymes; bioprocessing; plant-based; acetic fermentation.*

- SDG 2: Zero hunger; utilization of trimmings and by-products.
- SDG 9: Industry, innovation and infrastructure; development of waste treatment culinary technique.
- SDG 12: Responsible production and consumption; promotion of using the entirety of products.

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1. INTRODUCTION

This section of the report investigates the valorization of vegetables food waste within the context of fine-dining restaurants, followed by an overall view of enzyme mechanics and the potential of hydrolase-type enzymes to liquify plant tissue and structures. The final part of the section aims the exploration of acetic fermentation and vinegar production to transform the obtained product from the enzymatic reaction into a shelf stable culinary product.

1.1. Background

Geranium is a Danish fine-dining cuisine restaurant whose activity centers around a gourmet experience offer based on a degustation non modifiable menu. The project was developed in a collaboration between the student and the company and so for it, the history of the restaurant, storytelling, and relation with the topic are introduced.

1.1.1 History

The restaurant was founded the spring of 2007 by chef Rasmus Kofoed together with Søren Ledet, with financial support from Seier Capital. From the beginning, the ambition behind Geranium was to create a fine-dining restaurant rooted in Danish terroir, seasonality, and a holistic gastronomic experience that combines food, nature, and art.

Geranium initially opened in a different location from where it is situated today, specifically in Kongens Have (The King's Garden) in central Copenhagen. Three years later, the restaurant relocated and reopened in Parken Stadium in the Østerbro district, where it continues to operate today.

In the following years, the restaurant gained increasing international recognition. In February 2016, Geranium was awarded its third Michelin star, which still holds today. In 2022, the restaurant was ranked as the best restaurant in the world, further cementing its position as a leading figure in modern fine dining.

1.1.2 Storytelling

Geranium's early concept was based on the idea of creating a fine-dining restaurant focused on Danish terroir and seasonality. Over the years, this concept has evolved into the restaurant's current philosophy: a fully pescetarian and plant-focused cuisine that highlights freshness and the changing seasons of Denmark.

Both the dishes and the beverage offering are built around the use of fresh, seasonal ingredients sourced from organic local farmers as well as foraged products such as flowers and mushrooms. The menu is developed from the idea that the food presented on the plate should resemble a plant growing naturally out of the ceramic, reinforcing the strong connection between gastronomy and nature.

The beverage program follows the same philosophy, combining carefully selected wines and non-alcoholic pairings designed to complement menu and emphasize purity, balance, and Nordic flavors.

This sets the precedent for the upcoming development project, which, besides being scientifically focused, attends to the key ingredients for Geranium's formula.

1.1.3 Relation with the topic

Currently, the restaurant does not use enzymes in daily operations. The implementation of these as food aids comprehends a new path of research and development in the company. However, as the approach of enzymes use is from an environmental perspective to utilize raw vegetables food waste, that is a field that the restaurant does operate on.

The processes which Geranium implements to save food waste production are not based on using them as a source for subproduct development, but more regarding how to give a space in the dish to those parts of the product which are under-utilized. Additionally, another common practice is to use trimmings or parts from the raw product on staff food preparation.

According to vinegar usage, Geranium does not produce vinegar in-house. Despite that, acidity plays a structural role in its culinary identity and daily operations. The restaurant regularly produces pickled ingredients from seasonal herbs and flowers that are received in excess during peak season and later reintroduced into the menu. Examples include pickled pine, hops, elderflowers, and elderberries. These preparations are primarily used as seasoning elements, contributing brightness, aromatic lift, and balance to dishes and sauces.

In addition, several dishes throughout the restaurant's history have incorporated freshly pickled vegetables prepared on a weekly or daily basis, such as pickled kohlrabi or celeriac. These preparations demonstrate that controlled acidification is already integrated into the kitchen's workflow as a preservation and flavor-development technique.

Furthermore, Geranium relies on commercially sourced vinegars, such as cherry and elderflower vinegar, as tools for balancing preparations and rounding sauces, reductions, and final compositions. This indicates an established dependence on vinegar as a structuring and corrective element within the restaurant's flavor architecture.

Therefore, while Geranium does not currently produce vinegar internally, it operates within an acid-centered culinary framework that depends on vinegar for preservation, seasoning, and flavor modulation. The development of vegetable-derived vinegars from in-house by-products would not introduce a foreign technique but rather extend and internalize an already existing acidification culture, strengthening both ingredient circularity and conceptual coherence within the restaurant's gastronomic system.

1.1.4 Background synthesis

The background presented demonstrates a coherent alignment between Geranium's gastronomic identity and the proposed development project: the restaurant has structured its fine-dining model around Danish seasonality, terroir, and a holistic, plant-focused philosophy that integrates preservation, pickling, and controlled acidification as core operational tools. Nevertheless, while acidity already plays a structural role within its flavor architecture, the systematic application of enzymatic processes for transforming vegetables and valorizing plant-based by-products remains unexplored.

This operational gap, situated within an already sustainability-driven and research-oriented culinary framework, legitimizes the present initiative, which aims not to introduce an external paradigm but to develop integrated protocols for enzymatic transformation and selective fermentation capable of converting vegetable surplus into high-value gastronomic elements, thereby reinforcing circular resource management, technical innovation, and conceptual coherence within the restaurant's gastronomic system.

The key insights into the restaurant's background are publicly documented in these three core sources:

One Day at Geranium (Book): This definitive monograph documents the restaurant's daily rhythm and culinary philosophy, featuring high-art photography alongside Rasmus Kofoed's signature nature-inspired recipes (Christensen, Bent et al., 2017).

The World's 50 Best (Interview): This piece captures the team's reflections on reaching the global #1 spot and explains their 2022 pivot to a meatless menu focused on Scandinavian seafood and organic vegetables (*Geranium's Rasmus Kofoed and Søren Ledet Talk Birthday Celebrations and Permanent Plant-Based Pop-Ups*, 2025).

Identità Golose (Article): This professional profile provides a technical analysis of Geranium's evolution, specifically highlighting the synergy between their complex juice pairings and their award-winning 4,500-label wine cellar (Zanatta, Gabriele, 2022).

1.2. State of art

The following section compiles the scientific and technical knowledge relevant to this study and required for the development of the report. It reviews existing literature related to food waste (FW) treatment in gastronomy, enzyme-assisted plant tissue structural breakdown, temperature on flavor development and diffusion and acetic fermentation processes, considering both historical and contemporary applications.

1.2.1 Food Waste in Hospitality

1.2.1.1 The global scope of food waste

On a worldwide scale, a staggering one-third of all food produced for human consumption is ultimately lost or discarded, adding up to roughly 1.3 billion tons every single year (Ansari et al., 2024; Negrușa et al., 2025). Looking specifically at the European Union in 2022, statistics show that about 132 kilograms of FW were generated per inhabitant. A slim majority of this waste, at 54% or about 72 kilograms per person, came directly from residential households. The other 46% originated higher up the supply chain, which encompasses manufacturing, production, retail outlets, and restaurants.

Defining and targeting hospitality waste

When discussing FW within the context of the hospitality sector, the focus is generally on unwanted food that gets thrown away, such as meal preparation peels and leftovers scraped from guest plates (Papargyropoulou et al., 2019). However, researchers emphasize that this definition should intentionally leave out strictly non-edible byproducts that naturally occur during cooking and eating, like residual oils, bones, seeds, natural colorants, and flavorings (Filimonau & De Coteau, 2019). By removing these unavoidable items from the equation, the industry can focus its mitigation efforts squarely on avoidable FW (Papargyropoulou et al., 2019). This avoidable category accounts for an estimated 73-79% of the total volume of FW in the hospitality sector. Importantly, avoidable waste represents the only type of FW that service providers can partially or entirely control. This level of control exists because avoidable waste is primarily generated through inadequate techniques in preparation, storage, and transportation.

Sheer volume

FW constitutes a massive portion of the hospitality industry's overall refuse, carrying severe financial consequences for service providers. Within the European Union, the hospitality sector is responsible for approximately 12% of the total FW generated, placing it third behind households at 53% and the food processing industry at 19. This 12% figure is likely an underestimation, as it entirely excludes contract catering found in work and hospital canteens, as well as retail catering like supermarket cafes and coffee shops (Papargyropoulou et al., 2019).

To illustrate the historical scale of this issue, the EU hospitality sector generated over 12 million tons of FW in 2010. This problem is not confined to specific regions; Scandinavian operations discard over 0.45 million tons annually, significantly damaging both their corporate image and financial performance (Filimonau & De Coteau, 2019).

Empowering the workforce for change

Tackling this issue can have a noticeably positive impact on hospitality staff. Because employees interact directly with waste, they possess valuable firsthand knowledge regarding its character and quantity, allowing them to accurately identify the most critical operational areas for intervention. Furthermore, studies show that when staff members receive proper training and communication regarding the severe impacts of FW, they become far more likely to actively engage in waste minimization efforts and eagerly share this vital information with their customers (Filimonau & De Coteau, 2019).

1.2.2 Vegetable waste treatment technologies

Pretreatment mechanics

Before FW can be effectively processed into useful resources, it typically undergoes a pretreatment phase designed to accelerate the rate of hydrolysis, which ultimately makes the subsequent processing stages much more effective.

Physical and Chemical Pretreatment

Physical pretreatment methods heavily utilize various forms of radiation, including microwave, ultrasound, gamma, and electron beam radiation. In academic literature (Ansari et al., 2024), these physical techniques are usually categorized broadly into mechanical and thermal methods, although some researchers choose to classify them entirely separately from one another.

Conversely, chemical pretreatment remains one of the most widely adopted approaches. This category encompasses treatments using ionic liquids, carbon dioxide, oxidative agents, acidic solutions, and alkaline mixtures (Ansari et al., 2024). Typically, this involves applying specific acids like sulfuric or hydrochloric acid, or strong alkalis such as sodium hydroxide, potassium hydroxide, or ammonia. These robust chemical techniques are frequently deployed to prepare waste for hydrogen generation processes, anaerobic digestion, and anaerobic fermentation (Xiao et al., 2024).

Biological Pretreatment

Biological pretreatment presents a highly eco-friendly alternative that requires minimal energy and causes no negative environmental harm, though it does demand significant initial investment costs. This methodology relies on living biological agents to naturally break down FW and encourage enzymatic hydrolysis. Unlike their physical and chemical counterparts, biological methods completely bypass the need for harsh chemicals, extreme temperatures, or high-pressure environments (Xiao et al., 2024). The primary drawback of biological processing is that it is inherently much more time-consuming.

This biological category is split into three main classifications based on the active agent utilized (Ansari et al., 2024). Bacterial treatments employ species such as *Thermotoga*, *Bacillus*, and *Enterobacter*, and are incredibly common in the anaerobic digestion process. While bacterial methods successfully increase biofuel and hydrogen production without causing inhibitory effects, they are quite slow and do release greenhouse gases (Ansari et al., 2024).

Fungal treatments, utilizing species like *Monascus*, *Aspergillus*, and *Rhizopus*, are widely celebrated for their energy and cost efficiency during the hydrolysis of FW. Fungi easily break down complex organic molecules, carbohydrates, and proteins without any inhibition effects, but the process remains time-consuming and sometimes suffers from inconsistent efficiency (Ansari et al., 2024).

Finally, enzymatic treatments utilize commercial enzymes like amylase and protease to degrade the waste material. Similar to fungal applications, this approach breaks down complex organics and proteins in a thoroughly eco-friendly way, but its widespread adoption is heavily limited by the high purchasing cost of commercial enzymes and the need for large retention times (Ansari et al., 2024).

1.2.3 Vegetables tissue structures

Plants, such as vegetables and fruits are mostly water, however, all this liquid is retained by a variety of components and structures that are minority in the overall nutritional composition. These are starch (carbohydrates), dietary fiber and protein (Sagar et al., 2018).

1.2.3.1 Presence of dietary fiber in plants

Dietary fiber (DF) is considered an essential class of nutrient and defined as “the edible parts of plants or analogous carbohydrates that are resistant to digestion and absorption in the human small intestine with complete or partial fermentation in the large intestine” (DeVries, 2003).

Regarding the contents it is recognized that it includes a multiple number of different substances. These and the structure they form can be seen in Figure 1.

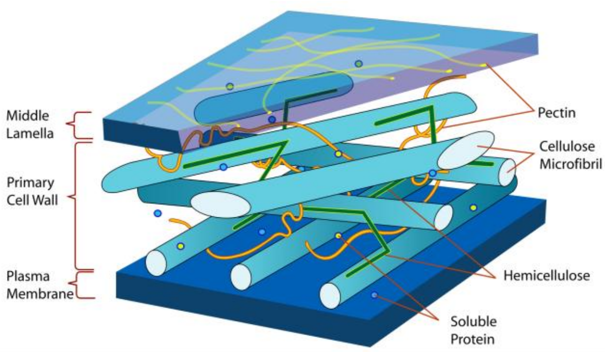

Figure 1
Plant cell tissue composition
(Source): (Regina, 2019).

Cellulose, which is the most abundant polysaccharide in the world and the most important part of the content regarding dietary fiber, and it is strictly formed of long chains of glucose. It represents approximately 20-45% DF content in vegetables and, as it is insoluble in water, it has the ability to bind it (Ozyurt & Ötles, 2016).

Hemicellulose, which is a heterogenous group like cellulose with the difference that its structure has monomer units other than glucose. It typically represents 15-35% of the DF in vegetables (Mudgil & Barak, 2013).

Pectin, which are polysaccharides found primarily in the plant cell walls, inter skin and rind of fruits and vegetables. Its content can fluctuate from 12-30% depending on the vegetable or fruit (Ridley et al., 2001).

Among these there is also the presence of hydrocolloids, beta-glucans and resistant starch, which are minority (DeVries, 2003).

1.2.3.2 Presence of starch in plants

Starch, is the primary energy storage polysaccharide in plants, composed of glucose polymers known as amylose and amylopectin. Unlike dietary fiber, most of the starch is highly digestible and is rapidly broken down and absorbed in the human small intestine. It traps water by reducing its mobility and, while not classified as DF itself, it is intrinsically linked to fiber within plant matrices (Tester et al., 2004).

1.2.4 Celery root composition

The main composition of celery root consists of: 88.7% of moisture, 4.3% available carbohydrate (from which 1.71g/100g are saccharose), 3.9% of dietary fiber, 1.8% protein and 0.3% fat (*Celeriac, Celery Root, Raw - Nutrition Data & Facts, Nutritional Information*, 2019).

According to Thimm et al (2002) , the primary cell walls of dicotyledonous plants typically contain 25–35% cellulose, 50% or more pectin with the reminder made up of xyloglucan and other hemicelluloses plus approximately 5% glycoproteins.

1.2.5 Enzymes

Enzymes are biological catalysts, primarily composed of proteins, that significantly accelerate the rate of specific chemical reactions within living organisms without being consumed in the process. They function by lowering the activation energy required for a reaction to occur, thereby facilitating vital metabolic processes, such as the breakdown of complex molecules into simpler ones (Robinson, 2015).

In a culinary and industrial context, they act as precision tools that can target and transform specific components of food. They contribute to more sustainable food processing methods by minimizing waste, reducing energy consumption, and decreasing reliance on chemical additives and drive reactions such as fermentation, texture modification, and flavor development by selectively binding to molecules in specific substrates (Paul et al., 2024).

1.2.5.1 Classification

In the food industry, enzymes can be divided into two main groups: endogenous and exogenous. Endogenous enzymes are naturally present in food tissues, whereas exogenous enzymes are intentionally added to raw materials to promote specific changes (Motta et al., 2023). Both groups can be classified into six main categories: oxidoreductases, transferases, hydrolases, lyases, isomerases, and ligases (Rigoldi et al., 2018). The only relevant for this report are hydrolases.

Hydrolases catalyze the cleavage of covalent bonds via hydrolysis, actively incorporating a water molecule into the reaction. This diverse group encompasses enzymes that break down carbohydrates (such as amylase, invertase, lactase, pectinase, xylanase, cellulase, and lysozyme), lipids (lipases), proteins (proteases), and nucleic acids (nucleases), with each type targeting a specific substrate (McDonald et al., 2009).

It is important to distinguish hydrolases from lyases. While lyases also break chemical bonds (such as pectin lyase and pectate lyase), they do so through elimination mechanisms rather than hydrolysis or redox reactions (Lara-Márquez et al., 2011).

1.2.5.2 Operating mechanism

According to Lutz et al (2012), there are three models that explain why enzymes have such high degree of specificity respectively to their operating mechanism. These are expressed in Figure 2.

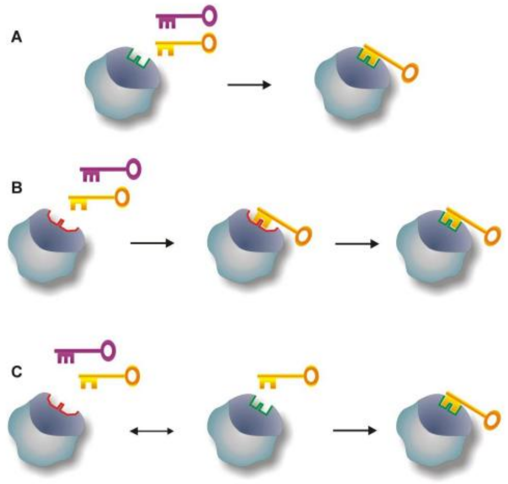

Figure 2

Enzyme catalysis models

Note: (A) *lock-key* model, (B) *induced-fit* model, (C) *selected-fit* model. (Source): (Lutz et al., 2012).

The lock-key model was proposed by Emil Fischer (1984), which compares the enzyme to a lock and the substrate to a key. In this model, the enzyme's active site has a fixed shape that is complementary to the substrate, allowing only properly matched molecules to bind.

In 1958, David Koshland introduced the induced-fit model, suggesting that enzymes are flexible and not all of them address rigid structures. According to this model, substrate binding induces a conformational change in the enzyme, allowing a better fit during catalysis.

The most modern explanation is the selected-fit model, which proposes that enzymes naturally exist in multiple conformations. The substrate binds to and stabilizes the most compatible conformation. Unlike induced fit, here the conformational change occurs before substrate binding (Vogt & Di Cera, 2012).

1.2.5.3 Enzyme activity respective to temperature

The catalysis activity of enzymes is affected by different parameters, from which one of the main is temperature. As described by Peterson et al. (2007), activity increases with temperature due to enhanced catalytic rates, but decreases at higher temperatures because of thermal denaturation. Enzymes can still function at temperatures below their optimum; however, their catalytic rate is significantly reduced. As a result, the time required to reach reaction endpoint becomes considerably longer at low temperatures, as illustrated in Figure 3.

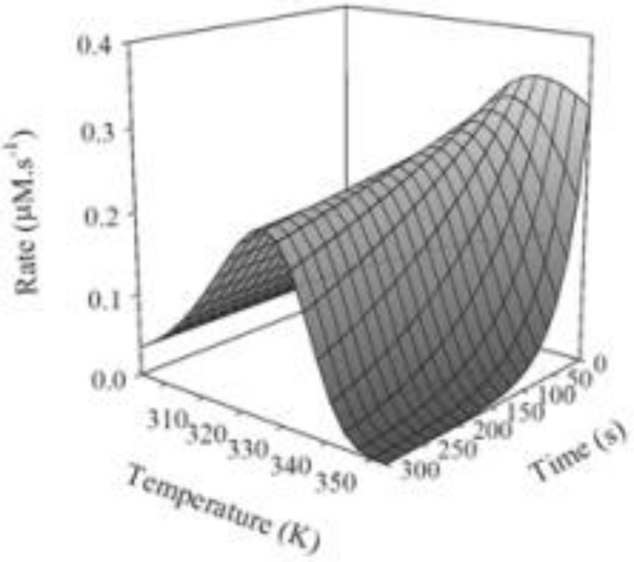

Figure 3

Temperature dependence, classical model

Note: the units used in the graphic are Kelvins with a conversion factor to celsius of K=°C+273.15.

(Source): (Peterson et al., 2007).

Although the curve in Figure 3 begins at 35 °C (310K), enzyme activity does not stop below this temperature. According to the Arrhenius relationship, lower temperatures reduce catalytic rate but do not eliminate activity. Therefore, enzymes can still function at refrigerator temperatures (~4 °C), despite reaction occurs way slower.

To explain how enzymes activity may decrease when lowering temperature below the optimal range, the Q10 coefficient can be used. This is a measure of temperature sensitivity that expresses how much the rate of a biological or chemical process changes with a 10 °C increase in temperature.

Traditionally, Q10 is derived from the Arrhenius equation, assuming reaction rates increase exponentially with temperature and often approximating that rates double for every 10 °C rise (Q10 ≈ 2). According to the coefficient calculation method provided by Mundim et al. (2020), it is possible to calculate the needed time for the enzymes to reach reaction completion at an unknown temperature with a known Q10 factor for the specific enzymes utilized. This can be done using Equation 1.

$$t2 = t1 * Q10^{\frac{T1-T2}{10}}$$

Equation 1

Time proportional enzyme activity based on the Arrhenius equation.

Note: t2=reaction time to calculate (min); t1=time at a known temperature (min); T2=new temperature to use (Cº); T1=temperature at a known time. (Source): adapted from (Mundim et al., 2020).

1.2.5.4 Enzyme kinetics and dosage

The amount of enzyme used in a process has a direct effect on the reaction rate. In general, when the enzyme concentration or the enzyme-substrate ratio increases, the reaction rate also increases because more active sites are available to catalyze the reaction. However, this increase is not unlimited. A point is eventually reached where adding more enzymes does not increase the reaction rate (Zhang et al., 2015).

This behavior can be explained using the Michaelis-Menten model of enzyme kinetics, represented in Figure 4.

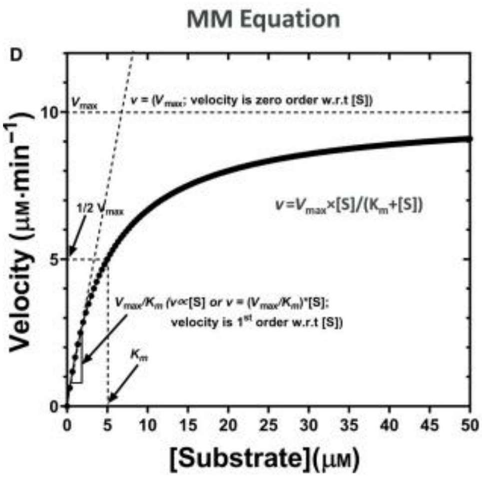

Figure 4

Michaelis-Menten model curve

Note: Michaelis–Menten kinetic curve showing the relationship between substrate concentration $[S]$and reaction velocity v. V_{max}represents the maximum reaction rate and $K_m$the substrate concentration at half V_{max}. (Source): (Srinivasan, 2022).

The graph shows the reaction velocity (v) as a function of substrate concentration ([S]) while the enzyme concentration ([E]) remains constant. At low substrate concentrations, the reaction rate increases rapidly because many enzyme active sites are available. As substrate concentration increases, the enzyme molecules gradually become saturated with substrate, and the reaction rate approaches a maximum value called Vmax (maximum velocity) (Srinivasan, 2022).

The value Km represents the substrate concentration required to reach half of the maximum velocity (Vmax/2). This parameter reflects the affinity between the enzyme and the substrate: a lower Km indicates a higher affinity (Srinivasan, 2022).

This model helps to explain why determining an optimal enzyme dosage is important. If too little enzyme is used, the reaction will proceed slowly. On the other hand, using excessive amounts of enzyme will not significantly increase the reaction rate once the system is close to Vmax, resulting in unnecessary enzyme consumption (Srinivasan, 2022).

1.2.5.5 Optimal conditions of use

The specific conditions of the enzymes used for the study are specified in 2.1.2 Methods and/or Annex 2, however, according to Wang et al (2023), for aqueous extraction food processing the maximum yield parameters for cellulase and pectinase are: pH 6, 80min, 50Cº. All these are coherent with the recommended conditions that are provided by Scientific and Technical Limited (*Enzyme Products for Industry, Developers, Researchers and Education*, 2026).

1.2.5.6 Legal frame regulation

The use of exogenous food enzymes in restaurant food preparation within Denmark is governed primarily by (Regulation (EC) No 1332/2008, 2008) on food enzymes, which establishes harmonized rules for the authorization and use of enzymes in foods across the European Union.

According to Article 3 of the Regulation, “food enzyme means a product obtained from plants, animals or micro-organisms or products thereof including a product obtained by a fermentation process using micro-organisms: (i) containing one or more enzymes capable of catalyzing a specific biochemical reaction; and (ii) added to food for a technological purpose at any stage of the manufacturing, processing, preparation, treatment, packaging, transport or storage of foods”. Under this definition, the use of exogenous enzymes in restaurant kitchens constitutes food preparation and therefore falls within the regulatory scope of the Regulation.

Article 4 stipulates that only authorized food enzymes included in the European Union Community list may be placed on the market and used in food production. Consequently, restaurants may legally use commercially available enzyme preparations that are approved for food use and marketed as food-grade materials. Responsibility for authorization lies primarily with the manufacturer or supplier, while food business operators such as restaurants must ensure that the enzyme is used in accordance with the supplier's specifications and applicable food safety requirements.

In addition, Article 6 establishes the general conditions for authorization and use of food enzymes, requiring that they do not suppose a risk to consumer health, that their use is technologically justified, and that they do not mislead the consumer.

When these conditions are fulfilled, the application of exogenous enzymes in restaurant food preparation is considered compliant with EU food law and is permitted within the Danish regulatory framework governing food business operators.

1.2.6 Plant hydrolyzing enzymes

This section focuses on the description of the 4 enzymes utilized in the project.

Cellulase enzyme

According to Ejaz et al. (2021), cellulase is a multicomponent enzyme complex responsible for degrading cellulose through hydrolysis of β-1,4-glycosidic bonds. Its mode of action relies on the synergistic activity of three main enzymes: endo-1.4-β-D-glucanase, exo1.4-β-D-glucanase and β-glucosidase. The process begins with endoglucanases, which randomly cleave internal β-1,4 linkages within the cellulose chain to increase substrate accessibility. Exoglucanases then act on the newly generated chain ends, progressively removing disaccharide units, mainly cellobiose, from the ends of the cellulose chains. The final step is catalyzed by β-glucosidase, which hydrolyzes cellobiose and short cello-oligosaccharides into glucose monomers. In summary, the mode of action of cellulase is a coordinated, multistep hydrolytic process: internal cleavage of cellulose chains, progressive removal of cellobiose units from chain ends, and final conversion to glucose (Horn et al., 2012; Lynd et al., 2002; Teugjas & Väljamäe, 2013).

Structurally, cellulases commonly consist of a catalytic domain linked to a cellulose-binding module (CBM). CBM enables the enzyme to bind tightly to insoluble cellulose surfaces, increasing local enzyme concentration at the substrate interface and enhancing catalytic efficiency. The catalytic domain performs hydrolysis through an acid–base mechanism, typically involving conserved glutamic acid residues that function as proton donor and nucleophile (Sweeney & Xu, 2012). This synergistic mechanism allows effective degradation of cellulose despite its crystalline structure and insolubility. Understanding this mechanism is essential when discussing dietary fiber modification, food processing applications, biomass conversion, and industrial saccharification processes.

Pectinase enzyme

Pectinases are a heterogeneous group of enzymes that act on the pectin molecules contained in the dietary fiber (DF) part of the plant. Pectins themselves are complex, high molecular weight, negatively charged acidic glycosidic polysaccharides. They account for 0.5%-4% of the fresh weight of plant material. In nature, these enzymes are naturally present in plants and help with relevant parts of growth, such as cell wall extension, ripening, and decomposition (Jayani et al., 2005).

Regarding their properties and mode of action, pectinase preparations, even when isolated, may contain the presence of glycosidases. Glycosidases are enzymes that hydrolyze glycosidic bonds, which release aglycons that were trapped in the structure attached to a sugar molecule and that act as an aromatic compound (Jayani et al., 2005). This is used for selective wine aroma enhancement, as done by (Rodríguez-Nogales et al., 2024).

Between all the industrial applications of pectinases, the main and most relevant for the project are applied for the fruit juice industry, wine industries, paper and pulp industries, and liquefaction of biomass. The fruit juice industry uses them to soften fruits, facilitate their extraction, and increase juice yield and clarification. In oil extraction industries, they are used to degrade the lamella in the cell walls together with the presence of other enzymes to improve the oil release from different sources (Shrestha et al., 2021).

Xylanase enzyme

Xylanase enzymes target xylan, which is the major component in hemicellulose and the polysaccharide with the most diverse structures in plant cells. Because xylan is highly branched and its structure can vary depending on the plant or substrate, a mix of enzymes is required to complete the hydrolysis of the structure (Polizeli et al., 2005).

Its mode of action relies on the cooperative activity of several specific enzymes: endoxylanase (endo-1,4-β-xylanase, E.C.3.2.1.8), β-xylosidase (xylan -1,4-β-xylosidase, E.C. 3.2.1.37), α-glucuronidase (α-glucosiduronase, E.C.3.2.1.139), α-arabinofuranosidase (α-Larabinofuranosidase, E.C. 3.2.1.55), acetylxylan esterase (E.C.3.1.1.72) (Polizeli et al., 2005). They all cooperatively transform the complex and branched structure of xylan into the sugars that conform it; however, among these, the most important is endoxylanase (Shahi et al., 2016).

Regarding the applications of xylanases, they are used in the paper industry, animal feed processing, clarification of beverages, and biofuel production. Xylanolytic enzymes (xylanases) are heavily used in the processing of juices and wines because they help break down structural polysaccharides in plant tissues. These enzymes make plant material softer and easier to process, which improves the extraction of liquids from fruits and vegetables. They are also used for oil extraction from plants and herbs. When combined with other enzymes such as cellulases, amylases, and pectinases, they help liquefy fruit and vegetable tissues more efficiently. As a result, the process increases the yield of valuable components such as aroma compounds, essential oils, vitamins, minerals, pigments, and natural colorants. The enzymes also break down substances that would otherwise interfere with clarification, preventing haze formation and improving the clarity and stability of the final juice or concentrate (Polizeli et al., 2005).

Amylase enzyme

Amylase is an enzyme that breaks down starch polymers into dextrin, maltose, and smaller glucose-formed chains by acting on the α–1,4 and α–1,6 molecule linkages (Mohanan & Satyanarayana, 2018). Regarding the industrial production of Amylases, the most used bacteria are *Bacillus* species such as *B. subtilis*, *B. licheniformis*, and *B. amyloliquefaciens*. *Aspergillus* filamentous fungi can also be used to produce amylase enzymes (Rehman & Saeed, 2015).

The hydrolysis of starch relies on the specific actions of three types of amylases:

- **α-Amylases:** These are endo-acting enzymes that hydrolyze internal α-1,4 glycosidic bonds into starch, producing maltose, maltotriose, and other oligosaccharides (Madhavi, n.d.).
- **β-Amylase:** This is an exo-acting enzyme that hydrolyzes α-1,4 glycosidic bonds from the non-reducing ends of starch, releasing maltose units as the primary product (Rani et al., n.d.).
- **γ-Amylases (glucoamylases):** These are exo-acting enzymes that hydrolyze α-1,4 and α-1,6 glycosidic linkages from the non-reducing end of starch, releasing free glucose units (Abd-Elhalem et al., 2015).

Another important factor about amylases is that their catalytic activity and structural stability are highly dependent on the presence of calcium ions (50-100 ppm for the one used in the project), which act as a necessary cofactor. Consequently, the mineral content of the water used in enzymatic extractions is a critical variable. Tap water often provides a sufficient natural source of calcium to maintain enzyme performance if specific mineral supplements are not added or naturally present in the substrate (Bush et al., 1989).

1.2.7 Temperature on flavor development and diffusion

Temperature significantly affects the flavor profile of fruit and vegetable juices through several physicochemical mechanisms. Thermal treatments (TT) commonly applied in juice processing, typically between 80 and 95 °C depending on the product and processing conditions, can induce multiple reactions that modify the final sensory characteristics of the product (Polak et al., 2024).

One important heat-induced reaction is the Maillard reaction, which occurs between reducing sugars and amino acids and generates aroma-active compounds such as diacetyl, furaneol, and heterocyclic compounds. Temperature also accelerates the oxidation of ascorbic acid, producing odor-active aldehydes. In addition, heat promotes the hydrolysis of flavor glycosides, releasing previously bound aroma compounds and increasing the concentration of volatile molecules (Zia et al., 2024)

According to Petruzzi et al (2017), the heat treatment of fruit and vegetables juices can have positive and negative effect on product quality. While certain processing conditions may enhance the formation of desirable flavor compounds, excessive temperature or prolonged treatment can deteriorate sensory quality. Particularly when treating with high temperature for long periods it reduces acceptance due to changes in the color and other attributes.

There's another branch regarding how temperature affects organoleptic profile. As reported by Daussy & Staudt (2020), when increasing temperature from 30ºC to 37ºC the diffusion/emission of volatile compounds from vegetable tissue triples according to the speed at lower temperatures.

Going deeper regarding color loss, temperature is critical as ranges of 60-80ºC begin the degradation of chlorophyll, and temperatures above 70ºC induce the degradation of carotenoids (Kardile et al., 2020; Manolopoulou & Varzakas, 2016).

As the optimal temperature for applying the enzymes is 50ºC (X. Wang et al., 2023), neither color loss or Maillard reaction should be reached, but there should be a considerable diffusion of volatile compounds from the vegetable tissue. On the other hand, doing the process below 30ºC may be suboptimal, but should be more respectful with the natural flavor of the product.

1.2.8 Fermentative process in vinegar production

Vinegar is a versatile, sour-tasting liquid primarily composed of water and acetic acid, which is created through a two-step natural fermentation process. First, yeast feeds on the natural sugars found in a carbohydrate source, such as apples, grapes, or grains, converting those sugars into alcohol. Next, specialized bacteria (known as *Acetobacter*) are introduced to ferment the alcohol into acetic acid, which gives vinegar its distinct pungent smell and tart flavor. Because of its unique acidic properties and wide range of flavor profiles depending on the source material, vinegar has been used for thousands of years across the globe as a cooking ingredient, a food preservative (pickling), and even as a natural household cleaner (El-Askri et al., 2022).

According to Regulation (EU) No. 1308/2013 (n.d.) international legal parameters, vinegar is strictly defined as a liquid fit for human consumption produced exclusively through a natural double fermentation process (first alcoholic, then acetous) originating from an agricultural carbohydrate source such as fruits, grains, or honey.

To legally qualify as vinegar, the liquid must meet mandated minimum acidity, ranging from 4% to 6% acetic acid, depending on the specific jurisdiction and the base material used. Furthermore, food and consumer protection laws require transparent labeling that clearly identifies the starting ingredient (e.g., "wine vinegar" or "malt vinegar") and strictly prohibit manufacturers from marketing diluted, chemically synthesized acetic acid as true, legally defined vinegar (n.d.).

Even though vinegar is traditionally and legally considered a two-step fermentation process, one is not dependent on the other, as they use different microorganisms and substrates. Rather, acetic fermentation is the consequence of alcoholic fermentation, as this one requires alcohol (Hua et al., 2024).

Alcoholic fermentation

Alcoholic fermentation is an anaerobic metabolic process where yeasts (such as *Saccharomyces cerevisiae*) and certain bacteria convert simple sugars into energy. The process begins in the cytoplasm of the cell with glycolysis, a sequence of reactions that breaks down one molecule of glucose into two molecules of pyruvate. This initial breakdown generates a net gain of two adenosine triphosphate (ATP) molecules, which the cell uses for energy, while also reducing the coenzyme *NAD+* into *NADH*. Because this metabolic pathway occurs entirely in the absence of oxygen, the cell cannot utilize the standard electron transport chain to regenerate the NAD+ required to keep glycolysis running. Therefore, it must rely on a different biochemical route to keep energy production from stalling (Walker & Stewart, 2016).

To sustain its energy output, the microorganism employs a precise, two-step enzymatic process to deal with the pyruvate. First, the enzyme pyruvate decarboxylase strips a carboxyl group from the pyruvate molecule. This reaction releases carbon dioxide as a metabolic waste product and leaves behind a highly reactive, two-carbon compound called acetaldehyde. Next, the enzyme alcohol dehydrogenase catalyzes the reduction of that acetaldehyde into ethanol. Crucially, this final step transfers hydrogen away from NADH, successfully oxidizing it back into its original NAD+ state (Rymuszka & Gorczynska, 2026).

This continuous recycling of NAD+ ensures that the yeast can survive and generate ATP indefinitely, even in completely anoxic environments. While this mechanism evolved purely as a microscopic survival strategy, humans have harnessed these exact metabolic byproducts for millennia. The carbon dioxide released during the decarboxylation step is what gives bread its airy rise and champagne its carbonation, while the resulting ethanol is the foundational alcohol in all beer, wine, and spirits and, as previously discussed, the essential precursor required to legally produce true vinegar (Rymuszka & Gorczynska, 2026).

Acetic fermentation

While alcoholic fermentation is an anaerobic survival tactic used by yeast, acetic fermentation is a strictly aerobic (oxygen-dependent) process driven by a completely different class of microorganisms: Acetic Acid Bacteria (AAB), most notably from the genus *Acetobacter*. In this phase, the bacteria do not break down complex sugars; instead, they feed directly on the ethanol produced during the first fermentation (El-Askri et al., 2022).

Because these bacteria obligately require oxygen to function, vinegar production cannot happen in a sealed, airtight vat. The liquid must be exposed to the air, either through the traditional, slow surface-level exposure (leaving barrels partially open) or through modern industrial submerged fermentation, where massive turbines constantly pump oxygen bubbles through the alcohol (Álvarez-Cáliz et al., 2020).

The biological transformation of alcohol into acetic acid occurs through a rapid, two-step incomplete oxidation process orchestrated by specialized enzymes located in the bacterial cell membrane. First, the enzyme alcohol dehydrogenase (ADH) interacts with the ethanol molecule, stripping hydrogen from it to produce acetaldehyde. Almost instantaneously, a second enzyme, aldehyde dehydrogenase (ALDH), captures this acetaldehyde, binds it with an oxygen molecule from the surrounding environment, and transforms it into acetic acid.

Unlike the carbon dioxide-heavy output of alcoholic fermentation, the only secondary byproduct of this specific chemical conversion is water (Hua et al., 2024).

The biochemical processes are shown in Figure 5.

$$\underset{\text{Ethyl alcohol oxygen}}{CH_3CH_2OH + (O)} \longrightarrow \underset{\text{Acetaldedyde Water}}{CH_3CHO + H_2O}$$

$$CH_3CHO + H_2O \longrightarrow \underset{\text{Hydrated acetaldehyde}}{CH_3CH + (OH)_2}$$

$$CH_3CH(OH)_2 + (O) \xrightarrow[\text{Dehydrogenase}]{\text{(Aldehyde)}} \underset{\text{Acetic acid}}{CH_3COOH + H_2O}$$

Figure 5

Acetic acid formation process

(Source): (Abdul Sattar, 2026)

In the figure can be seen how a molecule of ethanol plus an atom of oxygen ends up transforming into acetic acid with a 1:1 relation. This means that the percentage of alcohol that initially is, it will be the percentage of acetic acid in the final product, if all the ethanol has been transformed and if none of it has evaporated. From a practical point of view, and especially when making homemade vinegars, it could be said that the conversion will never be perfect, and so there will be a proportion lost in the process.

For this transformation to occur efficiently, the environmental parameters must be tightly controlled, balancing both the temperature and the presence of alcohol. The starting liquid must ideally contain an alcohol concentration between 5% and 12% by volume. If the alcohol level is above 15%, the environment becomes toxic for the AAB and will stall or die. On the other hand, being below 5% is not as critical. The main risk is for the process to find completion very early and then begin “over-oxidizing” the formed acetic acid, destroying the vinegar. However, if the fermentation process is forced to stop, this shouldn’t be a problem (Mota & Vilela, 2024).

There’s one more consideration regarding dropping below 5% initial alcohol, which is that, if as mentioned, by law a vinegar must have 4 to 6% acetic acid, the conversion is 1:1, and the transformation is unlikely perfectly efficient, doing a vinegar from a 3% alcohol won’t possibly develop enough acetic acid to be considered as such (Yin et al., 2018).

Temperature control is equally critical, and it happens similar as with the alcohol level, exceeding the limit from above is going to kill the reaction, but going below could be performed, just with some specific considerations (Hua et al., 2024).

The optimal operating range for AAB is between 25°C and 30°C, where the reaction will be performed the most efficiently. If the temperature drops below 20°C, the bacterial metabolism slows to a crawl, dragging a process that should take weeks into months. If the temperature exceeds 30°C, the bacteria become severely heat-stressed, and the highly volatile alcohol and newly formed acetic acid begin to evaporate into the air, resulting in a low-acidity, inferior final product. Above 42ºC the AAB will die. Therefore, mastering acetic fermentation requires a continuous, delicate balancing act of temperature mitigation, oxygen diffusion, and precise alcohol tracking (El-Askri et al., 2022). So, any procedure that can reach those temperatures long enough will pasteurize the vinegar and stop the fermentation process (Y. Wang et al., 2025).

1.3. Objectives

The present final degree project aims to develop and validate an enzymatic strategy for the value of raw vegetable food waste, enabling its transformation into vinegar within a professional restaurant context.

The main objective is to develop and evaluate a standardized enzymatic method for the liquefaction of raw and cooked vegetable waste. This within a four-month timeframe, determining its feasibility through the successful transformation of the resulting liquid substrate into a stable culinary vinegar adapted to the operational context of a professional kitchen.

In addition, and to reach the main goal, four secondary objectives were defined:

- Optimize the enzymatic reaction parameters to maximize liquid extraction yield while ensuring the process remains operationally practical for kitchen staff.
- Quantify the processing efficiency of the optimized method across distinct vegetable categories to establish reproducible conversion ratios and refine substrate-specific protocols.
- Execute a controlled acetic fermentation process to transform the vegetable extracts into common use culinary products.
- Validate the overall feasibility of the method by conducting a comparative analysis of economic costs and waste-reduction impact relative to the current resource management standards at Geranium

The limited scientific and technical knowledge regarding vinegar production from low-sugar vegetables, together with the growing need for sustainable, waste-based kitchen processes, justifies the realization of this study. The project seeks to establish a reproducible and operationally feasible method adapted to Geranium’s context, with potential applicability in other professional kitchens.

1.4. Phases of the project

To achieve the main and secondary objectives, the project is divided into five phases, considering that the present study will be carried out between February and early June. The work is structured into different phases to ensure proper planning, execution, and analysis of the results obtained. The phases of the project and their corresponding activities are described below.

Phase 1. Research and experimental design

- Review literature on vinegar production from vegetable food waste, enzymatic extraction, and fermentation processes.
- Select the vegetable substrates and define the experimental approach.
- Define experimental design, including enzymatic treatments and monitored parameters.

Phase 2. Enzymatic treatment evaluation

- Apply selected enzymatic treatments to define the parameters for a vegetable liquefication method.
- Measure extraction yield (%).
- Evaluate the flavor potential of the obtained liquid.

Phase 3. Evaluation on different substrates

- Test the optimized methodology in a variety of products that are part of the restaurant's daily operations.
- Fix the parameters for those products that work inefficiently.
- Select the most suitable substrates for vinegar production.

Phase 4. Fermentation performance assessment

- Monitor fermentation performance (pH evolution, acetification time, and stability).
- Build a procedure for pasteurization and fermentation stopping.
- Create recipes that are appliable in the restaurant.

Phase 5. Results and feasibility

- Analyze the technical results obtained during the experimental phase.
- Quantify the environmental impact based on vegetable waste utilization.
- Quantify the economic potential.
- Define the conditions for implementing the optimized process in a restaurant context.

The phases described above are organized in a project schedule represented by a Gantt chart, included in Annex 1. This planning tool allows a clear visualization of the project timeline and supports the structured management of all stages of the study.

2. DEVELOPMENT

2.1. Materials and methods

2.1.1. Materials

All the materials, utensils and products utilized for the development of the project are summarized in a three columns table (item – description – figure) that can be seen in Annex 2 (Table 38).

2.1.2. Methods

The next section determines the practical information and methodology required for the development section and experimental designs.

Enzymatic reaction process optimization

The first stage of the development started with building a method and optimizing it for the chosen testing substrate (celery root pomace). The procedure was the following:

On one hand, the conditions and parameters for the reaction were chosen based on Table 1, that summarizes the specifications given by the isolated enzymes provided by Scientific and Technical Limited.

Table 1

Enzyme utilization parameters

	Cellulase	Pectinase	Amylase	Xylanase
pH Range	3.5 - 6.0	3.5 - 8.0	5.0 - 8.0	3.4 - 8.0
pH Optimal	4.5 - 5.0	5.8 - 6.5	5.8 - 6.2	4.5 - 5.5
Tº Range (Cº)	30 - 60	15 - 60	40 - 105	15 - 55
Tº Optimal (Cº)	50 - 55	45 - 60	85 - 95	50 - 55
Dosage	0.05% - 0.2%	0.1% - 0.5%	0.05% - 0.1%	0.05% - 0.2%

(Source): (*Enzyme Products for Industry, Developers, Researchers and Education*, 2026).

The data was obtained from the technical datasheets that can be seen in Annex 3. From this, the conditions chosen: temperature of 50ºC; pH of 5.5; cellulase dosage of 0.2; pectinase dosage of 0.4; and amylase dosage of 0.5. The amylase dosage chosen was large with the intention to compensate that it was acting far from optimal temperature range. Xylanase was not introduced as it was understood as a secondary enzyme.

On the other hand, a procedure was built for testing the enzymatic reaction. This can be seen in the diagram shown in Figure 6.

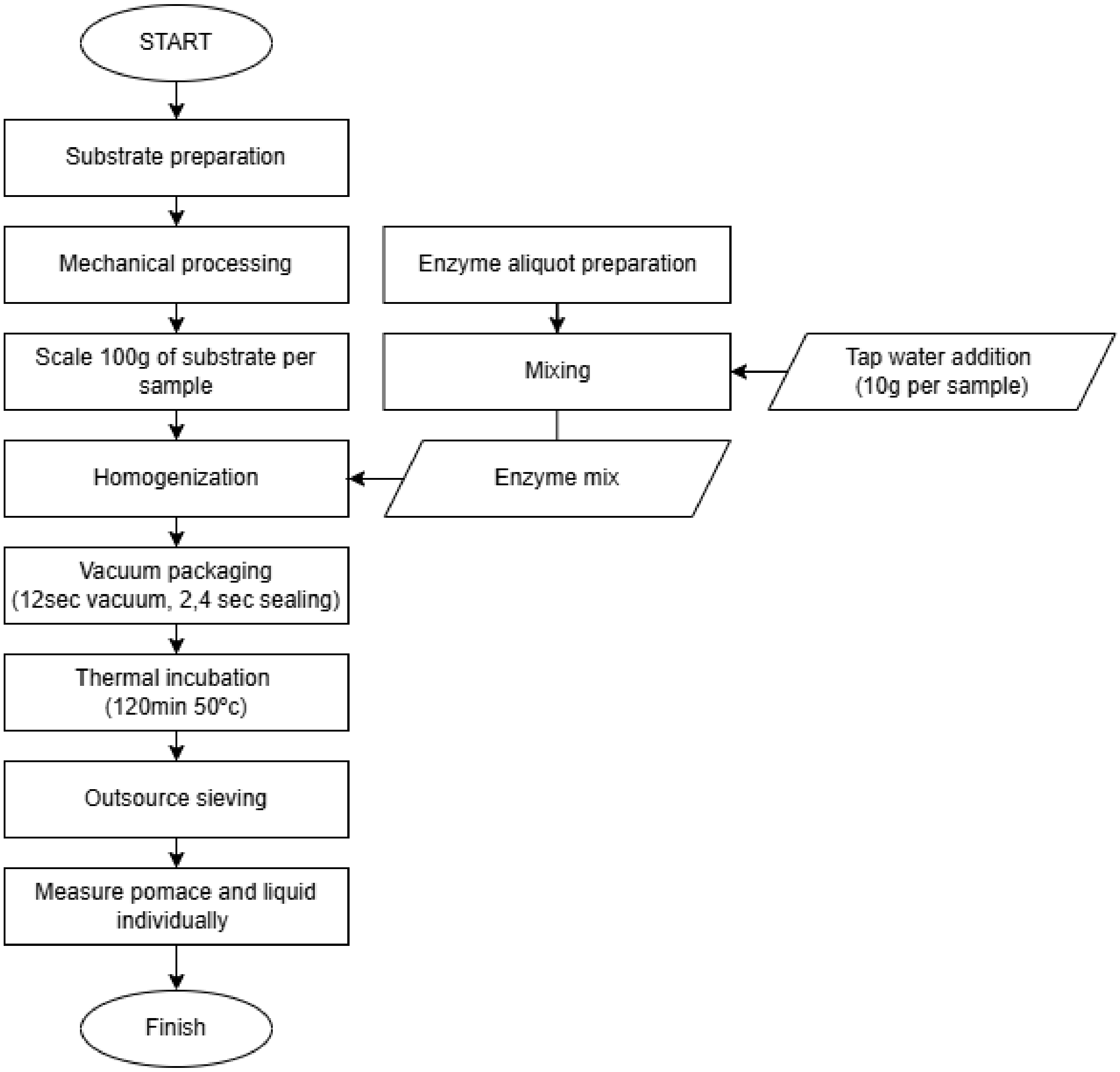

Figure 6

Enzyme practical testing method diagram

First, the “substrate preparation”. Which varies depending on the substrate utilized and the development section, this includes tasks such as peeling, washing, coring trimming and cooking. However, for the first tests and optimization the only thing that was done to the celeriac was peeling.

Besides the first step, the rest of the procedure always remains the same (unless it is specifically mentioned) and breaks down as follows:

“Mechanical processing”, transforming the whole product either by mechanically blitzing with a Thermomix or chopping down with a knife.

The second part is to scale a 100g sample of the processed products and mix it with the required enzymes dosage and a small part of water (10%).

This is straight transferred into bags and vacuumed for 12 seconds with 2.4 seconds sealing, what provides an environment with enough pressure so there is not space between the product and the bag, and the thermal transference is effective, but not as much as so the substrate cannot be mixed inside the bag.

The sealed samples go into a water bath at 50ºC for 120 minutes for the reaction to happen and it is finished by thieving, scaling and results annotation.

Finally, to test and improve the method, four different experiments were performed:

The first one focused on applying the reaction on 6 different samples the same exact way but stopping and measuring the obtained yield at different times. This is for understanding how long it takes until completion and how the obtained yield evolves throughout time.

The second one, to test the different enzymes individually and in pairs to evaluate how they perform by separate and the importance of using them all together.

The third one, to evaluate how significant is the addition of liquid or water for the reaction to perform efficiently and if the natural liquid content of the vegetables is enough.

The last one, with the conditions optimized, reproduces the method but at refrigeration temperature (4ºC), to see if the reaction can happen in the fridge. In this, the time efficiency, completion time and yield were evaluated to compare them with the first one.

To set a theoretical time for scaling the reaction to refrigeration temperature, Equation 1 was utilized. Using as reference a Q10 factor of 1.53 for pectinase at 60ºC for 169 minutes, which is the lowest acting enzyme (Trindade et al., 2016), it was estimated that the reaction endpoint would be reached after 30.5 hours.

Method appliance on different substrates

With a built working procedure to perform the enzymatic reaction, the second phase focuses on the evaluation of the same procedure on a variety of different products.

The mechanism to evaluate substrates remained the same as the one shown on Figure 6. And the tests performed are the next:

Number one, it consisted of trying to find out if the same that had already worked on celery root pomace would on other products and, if so, if it would remain as effective. For this, seven ingredients were selected out of a list (Table 2) and chosen by Ronni Mortensen based on presence in the menu and coherence with the storytelling.

Table 2

Ingredients selection list

Vegetables	Herbs	Fruit
Potato	Mizuna	Lemon
Beetroot	Lemon verbena	Apple
Pumpkin	Chervil	Cucumber
Cauliflower	Peashoot	
Flower sprout	Thyme	
Broccolini	Tarragon	
Carrot	Watercress	
Kholrabi	Nasturgium	

Note: the products highlighted in yellow are those which were chosen.

The table was built with the ingredients available in the vegetable's fridge that same day. Those highlighted yellow are the chosen.

Number two, it followed the same system as the previous, but the products chosen were chosen based on botanical family specifically based on the table that can be found in Annex 4. These are: leek (Amaryllidaceae), fennel (Apiaceae), pepper (Solanaceae), zucchini (Cucurbitaceae), cucumber (Cucurbitaceae), pumpkin (Cucurbitaceae) and beetroot (Amaranthaceae), which was done before, but this time it was cooked to prove that is a requirement for that botanical family. These complement the previous botanical families tried: Brassicaceae, Asteraceae, Apiaceae, Solanaceae, Amaranthaceae.

Number three, focused on proving that Brassicaceae family requires to be cooked and cannot be liquified when raw. Aswell, a last chance was given to herbs. The experiment consisted of broccolini and kohlrabi both when raw and cooked. Tarragon was used as the herb and a sample of jerusalem artichoke was added raw to prove that tarragon could not be liquified due to that is an herb and not because it does not work for the Asteraceae family.

Number four, and last, consisted of testing if fine mechanical processing provides higher yield than using bigger pieces or chunks. For this, three ingredients were tested beetroot, jerusalem artichoke and potato.

Substrate selection for the vinegars

Already having built knowledge of the flavor profile of a variety of raw ingredients enzymatic juice, the method used to try out new flavors and choose which would be the candidates to develop the vinegars with focused on previously cooked products.

To choose which would be the ingredients and cooking method applied for the testing it was not used a specific procedure, but rather knowledge of the restaurants operations to select ingredients that are signature for the flavor structure of the restaurant and cooked with a method that is simple and can be applied regardless of shape (so it can be applied to food trimmings).

The selected products: potato, boiled and roasted with and without skin; red bell pepper, roasted; beetroot, boiled with and without skin; garlic pulp, roasted; quince, roasted; pumpkin, cored and roasted; celeriac, roasted; jerusalem artichoke, roasted whole and peeled. The exact data regarding preparation process is included in 2.2.3 Phase 3: Substrate selection for the vinegars.

With the juices obtained after the liquefication process an informal tasting was performed to select 5 to be used. That together with feedback from Ronni Mortensen on how to improve the overall flavor. Information utilized to do a second try with changes in the recipe such as fat utilization, water content and cooking time.

Acetic fermentation method

The production method for the acetic fermentation can be seen on Figure 7.

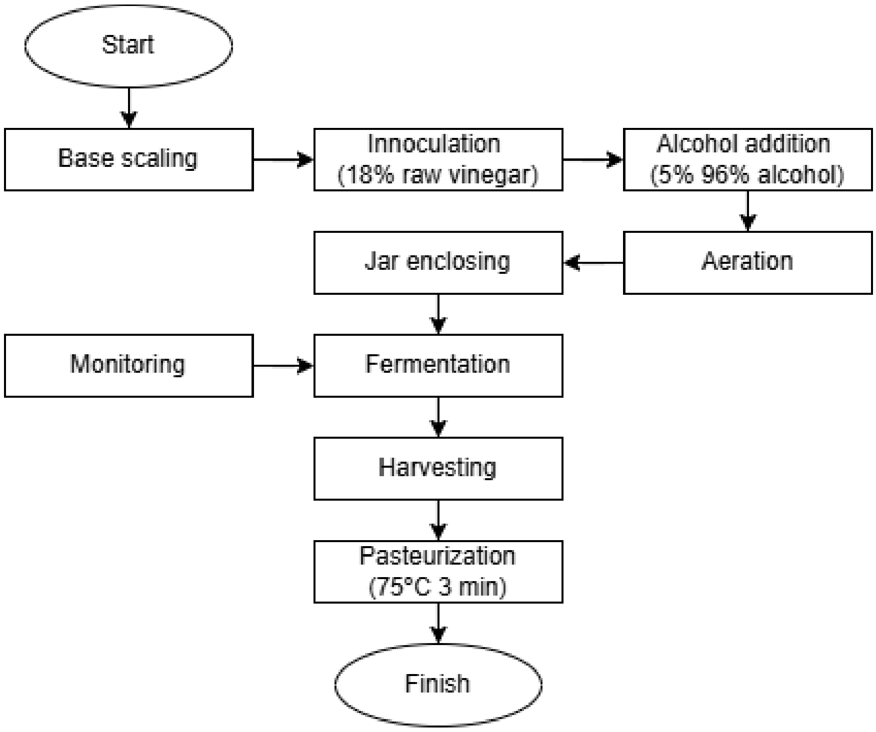

Figure 7

Acetic fermentation diagram

The first step to produce the vinegars was adding a fermentation starter into the base liquid. For this, a raw unfiltered apple cider vinegar with a 5% acetic acid was added into each of the base juices/liquids, exactly an 18% over the weight of base juice. It was done for 1 kg of base, mixed with a sanitized whisk and poured into a 3L volume glass jar.

With the use of a pH meter, it was checked that every sample was below 4.5 ph. Over the whole weight of liquid, it was added and whisked in a 5% of 96% distilled alcohol for food use.

An air stone attached at the end of an air pump with a maximum capacity of 4.4L of air per minute was introduced in the liquid and set at the center bottom of the jar to provide continuous oxygenation. For every two jars, one air pumping machine was used with the regulator set at the lowest speed. With the machines working all the jars were fixed with breathable kitchen cloth and a rubber band.

From then, every day at 17:00 a 20ml sample was picked out of each jar with a sterilized ladle to check pH, temperature and Brixº. Added on, each sample was tasted and the remaining liquid returned into the jar.

Once the fermentation process was finished, and so the pH level stable, 800g of each were poured into 1L volume bottles and immersed in a water bath at 80ºC for 5min. Right after immediately transferred to an ice bath where to quickly drop the temperature.

2.2. Development

This section focuses on all the tests preparations for the enzymatic reaction development and optimization, the testing on different products, the production of vinegar and further product validation systems.

2.2.1. Phase 1: Enzymatic reaction process optimization

Enzyme reaction time evaluation

To evaluate how the enzymes perform throughout time a test was built with six different samples. The utilized substrate was celeriac pomace, and the procedure applied is the following:

The celeriac pomace as a leftover from the restaurant's juice production, for which was peeled and processed in a juicing machine, it was utilized 200g per sample mixed with another 200g of tap water and the enzymes dosage to the weight of celeriac. The pH was corrected to 5.5 with the addition of 0.25% (w/w) ascorbic acid. Every sample was transferred into a vacuum bag and vacuumed for 12 seconds and sealed for 2.4 seconds. With all of them ready, they were introduced simultaneously in a water bath at 50ºC for different amounts of time. After reaction time completed, they were all strained through a fine mesh chinois and both the pomace left and the juice obtained were scaled by separate.

In Table 3 can be seen the specific parameters of each sample.

Table 3

Enzyme reaction time

	Reaction Parameters					Dosage %		
	Substrate	T(°C)	Regime	Time uts	pH	Cellulase	Pectinase	Amylase
1	Celeriac pomace +100% water	50	*Sous-vide*	150 min	5.5	0	0	0
2	Celeriac pomace +100% water	50	*Sous-vide*	30 min	5.5	0.2	0.4	0.5
3	Celeriac pomace +100% water	50	*Sous-vide*	60 min	5.5	0.2	0.4	0.5
4	Celeriac pomace +100% water	50	*Sous-vide*	90 min	5.5	0.2	0.4	0.5
5	Celeriac pomace +100% water	50	*Sous-vide*	120 min	5.5	0.2	0.4	0.5
6	Celeriac pomace +100% water	50	*Sous-vide*	150 min	5.5	0.2	0.4	0.5

Note: sample number 1 does not have enzymes as it was used as control.

Enzyme activity testing

As the method proposed in the previous experiment worked efficiently and provided a highly noticeable yield, the second one imitates the set conditions. The aim this time is to understand which of the enzymes are providing more yield individually and in combination.

The specific parameters set for the experiment can be seen in Table 4.

Table 4

Enzyme activity testing

	Reaction Parameters					Dosage %		
	Substrate	T(°C)	Regime	Time uts	pH	Cellulase	Pectinase	Amylase
1	Celeriac pomace +100% water	50	*Sous-vide*	120 min	5.5	0	0.4	0
2	Celeriac pomace +100% water	50	*Sous-vide*	120 min	5.5	0.2	0	0
3	Celeriac pomace +100% water	50	*Sous-vide*	120 min	5.5	0	0	0.5
4	Celeriac pomace +100% water	50	*Sous-vide*	120 min	5.5	0.2	0.4	0
5	Celeriac pomace +100% water	50	*Sous-vide*	120 min	5.5	0	0.4	0.5

Liquid-solid interaction and pH

Based on the conclusions drawn from the results of the previous testing, an experiment was built to replicate the process while altering the amount of added liquid. Additionally, the pH was not fixed to 5.5 but left to the natural level of juiced celeriac pomace (around 6). The specific parameters are detailed in Table 5.

Table 5

Liquid-solid interaction and pH

	Reaction Parameters					Dosage %		
	Substrate	T(°C)	Regime	Time uts	pH	Cellulase	Pectinase	Amylase
1	Celeriac pomace	50	*Sous-vide*	120 min	6.2	0.2	0.4	0.5
2	Celeriac pomace + 30% juice	50	*Sous-vide*	120 min	6.2	0.2	0.4	0.5
3	Celeriac pomace + 60% juice	50	*Sous-vide*	120 min	6.2	0.2	0.4	0.5
4	Celeriac pomace + 100% juice	50	*Sous-vide*	120 min	6.2	0.2	0.4	0.5
5	Celeriac pomace + 30% water	50	*Sous-vide*	120 min	6.2	0.2	0.4	0.5

Refrigeration temperature testing

The last of the experiments respective to enzymatic reaction process optimization aims to prove if the method proposed can be applied at refrigeration temperature. The details can be seen in Table 6. The procedure was the exact same but happened in the fridge rather than in a water bath.

Table 6

Refrigeration temperature testing

Reaction parameters						Dosage %		
	Substrate	T(°C)	Regime	Time uts	pH	Cellulase	Pectinase	Amylase
1	Celeriac pomace	4	Refrigerator	12 h	6.2	0.2	0.4	0.5
2	Celeriac pomace	4	Refrigerator	18 h	6.2	0.2	0.4	0.5
3	Celeriac pomace	4	Refrigerator	24 h	6.2	0.2	0.4	0.5
4	Celeriac pomace	4	Refrigerator	36 h	6.2	0.2	0.4	0.5

2.2.2. Phase 2: Method appliance on different substrates

This section evaluates the performance of the optimized enzymatic reaction method from the previous section on a variety of vegetables.

Evaluation on raw generic vegetables

The first experiment regarding evaluation of different substrates copies the parameters utilized and method previously set. The specific parameters can be seen in Table 7 together with the chosen substrates.

Table 7

Evaluation on raw generic vegetables

Reaction Parameters							Dosage %		
	Substrate		T(°C)	Regime	Time uts	pH	Cellulase	Pectinase	Amylase
1	Carrot	Apiaceae	50	*Sous-vide*	120 min	5.5-6.4	0.2	0.4	0.5
2	Potato	Solanaceae	50	*Sous-vide*	120 min	5.5-6.4	0.2	0.4	0.5
3	Beetroot	Amaranthaceae	50	*Sous-vide*	120 min	5.5-6.4	0.2	0.4	0.5
4	Cauliflower	Brassicaceae	50	*Sous-vide*	120 min	5.5-6.4	0.2	0.4	0.5
5	Broccolini	Brassicaceae	50	*Sous-vide*	120 min	5.5-6.4	0.2	0.4	0.5
6	Kohlrabi	Brassicaceae	50	*Sous-vide*	120 min	5.5-6.4	0.2	0.4	0.5
7	Tarragon	Asteraceae	50	*Sous-vide*	120 min	5.5-6.4	0.2	0.4	0.5
8	Lemon Verbena	Verbenaceae	50	*Sous-vide*	120 min	5.5-6.4	0.2	0.4	0.5

Note: the specific pH can vary depending on the natural acidity of the substrate.

Regarding enzyme application, the same combination was used for all samples. Although certain enzymes were not theoretically expected to catalyze reactions with specific substrates (e.g., amylases with tarragon), it was decided to use the entire mix combination to maintain uniform conditions across the study.

Additionally, as the substrates used are not leftovers from previous restaurant's processing (like it was the celeriac pomace), it was finely chopped into a 2 mm x 2 mm *brunoise*. This decision was also intended to minimize experimental deviation, as reduced processing and fewer vessel transfers decrease waste and lead to more accurate results.

Vegetables by botanical family

As concluded in the previous experiment, before further method changes, it was required to study vegetables from other botanical families. For this, Table 8 was drawn with the conditions and selected products.

Table 8

Vegetables by botanical family

Reaction Parameters						Dosage %		
Substrate		T(°C)	Regime	Time uts	pH	Cellulase	Pectinase	Amylase
1 Leek	Amaryllidaceae	50	*Sous-vide*	120 min	5.5-6.4	0.2	0.4	0.5
2 Fennel	Apiaceae	50	*Sous-vide*	120 min	5.5-6.4	0.2	0.4	0.5
3 Red bell pepper	Solanaceae	50	*Sous-vide*	120 min	5.5-6.4	0.2	0.4	0.5
4 Zucchini	Cucurbitaceae	50	*Sous-vide*	120 min	5.5-6.4	0.2	0.4	0.5
5 Cucumber	Cucurbitaceae	50	*Sous-vide*	120 min	5.5-6.4	0.2	0.4	0.5
6 Pumpkin	Cucurbitaceae	50	*Sous-vide*	120 min	5.5-6.4	0.2	0.4	0.5
7 Jerusalem artichoke	Asteraceae	50	*Sous-vide*	120 min	5.5-6.4	0.2	0.4	0.5

Since it was established that Brassicaceae possess a hemicellulose wall that impedes substrate permeability, other products were tested to verify that different botanical families could be liquefied using the developed method before applying xylanase to resolve the reaction.

Vegetables when cooked and mechanically processed

As cooking softens internal structure and molecule bindings, which makes the substrate easier to degrade and increases the permeability of the product, what allows the specific enzymes to contact the substrate they catalyze, it should be a way to improve yield (Gonzalez & Barrett, 2010). Further increased by the mechanical blitzing, that spreads the enzyme within the mixture and makes the pieces of product smaller for the liquefaction process. To understand how these may affect the substrate before the enzymatic reaction, the experiment in Table 9 was drawn.

Table 9

Vegetables when cooked and mechanically processed

Reaction Parameters						Dosage %		
Substrate	T(°C)	Regime	Time	uts	pH	Cellulase	Pectinase	Amylase
1 Raw chopped jerusalem artichoke	50	*Sous-vide*	120	min	5.5-6.4	0.2	0.4	0.5
2 Boiled chopped jerusalem artichoke	50	*Sous-vide*	120	min	5.5-6.4	0.2	0.4	0.5
3 Boiled jerusalem artichoke purée	50	*Sous-vide*	120	min	5.5-6.4	0.2	0.4	0.5
4 Raw chopped potato	50	*Sous-vide*	120	min	5.5-6.4	0.2	0.4	0.5
5 Boiled chopped potato	50	*Sous-vide*	120	min	5.5-6.4	0.2	0.4	0.5
6 Boiled potato purée	50	*Sous-vide*	120	min	5.5-6.4	0.2	0.4	0.5

Sample 1 and 4 were recycled from previous experiments. Regarding the boiling of the roots, it was performed at a simmering temperature (90-95ºC) in triple the amount of water than product during 30 minutes for the jerusalem artichoke, and 45 for the potatoes. Then they were peeled when still warm, a part chopped by knife until 2x2mm pieces and the other part processed in a Thermomix at 5.5 speed for 50 seconds.

Brassicaceae family

The experiment details can be seen in Table 10.

Table 10

Brassicaceae family

Reaction Parameters						Dosage %			
Substrate	T(°C)	Regime	Time	uts	pH	Cellulase	Pectinase	Amylase	Xylanase
1 Raw chopped broccolini	50	*Sous-vide*	120	min	5.5-6.4	0.2	0.4	0.5	0.1
2 Cooked chopped broccolini	50	*Sous-vide*	120	min	5.5-6.4	0.2	0.4	0.5	0.1
3 Raw chopped kohlrabi	50	*Sous-vide*	120	min	5.5-6.4	0.2	0.4	0.5	0.1
4 Cooked chopped kohlrabi	50	*Sous-vide*	120	min	5.5-6.4	0.2	0.4	0.5	0.1

The focus of the experiment is to try and make the method work on Brassicaceae botanical family. The first proposed solution explained was the implementation of Xylanase. So, a dosage of 0.1% was added.

The second proposed solution is to pre-cook the product. This approach is based on the fact that Brassicaceae naturally contain myrosinase, a hydrolase enzyme that converts glucosinolates into isothiocyanates. These compounds can inactivate the exogenously added enzymes, thereby preventing the intended reaction. Heat treatment effectively denatures the myrosinase, mitigating this interference (Barba et al., 2016).

To perform the experiment, broccolini and kohlrabi (previously peeled and chopped into 2cm wide chunks) were cooked at a simmering temperature (90-95ºC) in triple the amount of water than product during 3 minutes for the broccolini, and 13 for the Kohlrabi. Then a part was chopped by knife into 2x2mm pieces and the other part processed in a Thermomix at 5.5 speed for 50 seconds.

Amaranthaceae family

The experiment parameters can be seen in Table 11.

Table 11

Amaranthaceae family

Reaction Parameters						Dosage %			
	Substrate	T(°C)	Regime	Time uts	pH	Cellulase	Pectinase	Amylase	Xylanase
1	Raw chopped beetroot	50	*Sous-vide*	120 min	5.5-6.4	0.2	0.4	0.5	0.1
2	Boiled chopped beetroot	50	*Sous-vide*	120 min	5.5-6.4	0.2	0.4	0.5	0.1
3	Raw beetroot purée	50	*Sous-vide*	120 min	5.5-6.4	0.2	0.4	0.5	0.1
4	Cooked beetroot purée	50	*Sous-vide*	120 min	5.5-6.4	0.2	0.4	0.5	0.1

The aim of the experiment is to evaluate the yield difference of beetroot when raw and chopped, boiled and chopped, raw purée, and purée when cooked. The chopped samples were done by knife into 1x1mm cubes, the purées processed in a Thermomix for 1 min at full speed and the cooking method applied 2 hours simmering from cold water.

Herbs

The details of the experiment can be seen in Table 12.

Table 12

Herbs

Reaction Parameters						Dosage %			
	Substrate	T(°C)	Regime	Time uts	pH	Cellulase	Pectinase	Amylase	Xylanase
1	Raw chopped tarragon	50	*Sous-vide*	120 min	5.5-6.4	0.2	0.4	0.5	
2	Raw chopped tarragon + Xyl	50	*Sous-vide*	120 min	5.5-6.4	0.2	0.4	0.5	0.1
3	Blanched chopped tarragon	50	*Sous-vide*	120 min	5.5-6.4	0.2	0.4	0.5	0.1

Three samples of tarragon were tested in this experiment with the purpose of finding a way to make the method work on herbs. The first sample chopped fresh tarragon with the basic enzymes mix, the second with added Xylanase to see if it would make the difference and the third previously blanched for 1 minute in boiling water and immediately transferred to an ice bath.

2.2.3. Phase 3: Substrate selection for the vinegars

First substrate selection approach

All the substrates and the cooking/preparation method are enlisted in Table 13.

Table 13

First substrate selection approach

	Reaction Parameters						Dosage %			
	Substrate	Water	T(°C)	Regime	Time uts	pH	Cellulase	Pectinase	Amylase	Xylanase
1	Potato whole boiled and roasted	10%	50	*Sous-vide*	120 min	5.5-6.4	0.2	0.2	0.1	0.1
2	Potato peeled boiled and roasted	10%	50	*Sous-vide*	120 min	5.5-6.4	0.2	0.2	0.1	0.1
3	Red bell pepper roasted	10%	50	*Sous-vide*	120 min	5.5-6.4	0.2	0.2	0.1	0.1
4	Beets boiled peeled	10%	50	*Sous-vide*	120 min	5.5-6.4	0.2	0.2	0.1	0.1
5	Beets boiled whole	10%	50	*Sous-vide*	120 min	5.5-6.4	0.2	0.2	0.1	0.1
6	Garlic roasted pulp	10%	50	*Sous-vide*	120 min	5.5-6.4	0.2	0.2	0.1	0.1
7	Quince cored roasted	10%	50	*Sous-vide*	120 min	5.5-6.4	0.2	0.2	0.1	0.1
8	Pumpkin cored roasted	10%	50	*Sous-vide*	120 min	5.5-6.4	0.2	0.2	0.1	0.1
9	Celeriac peeled roasted	10%	50	*Sous-vide*	120 min	5.5-6.4	0.2	0.2	0.1	0.1
10	Jerusalem Artichoke whole and roasted	10%	50	*Sous-vide*	120 min	5.5-6.4	0.2	0.2	0.1	0.1
11	Jerusalem Artichoke peeled roasted	10%	50	*Sous-vide*	120 min	5.5-6.4	0.2	0.2	0.1	0.1

The test focused only on cooked vegetables, as the raw flavor of the chosen products had already been tasted. Most of them are roasted as it is a cooking method that can give depth and improve the flavor overall, furthermore it can be easily standardized and equally used both on whole product or trimmings.

Eight products were tested: potato, red bell pepper, beetroot, garlic, quince, pumpkin, celeriac and jerusalem artichoke. Besides the beetroot, that was simmered for 50 minutes, they all were roasted at 195ºC chopped into big chunks until having a considerable amount of Maillard reaction, which means a time of 25 to 40 minutes.

With potato, beetroot and jerusalem artichoke two samples were done, with and without skin. Also, potato and jerusalem artichoke were simmered for 25 minutes in water before roasting.

Regarding the enzymatic reaction, the dosage of enzymes was reduced: for amylase from 0.5% to 0.1%; for pectinase, from 0.4% to 0.2%. With the intention of reducing enzyme requirements before applying to bigger batches, and still in the optimal utilization parameters.

Chosen substrates cooking improvement

Based on the conclusions and results obtained from the previous product tasting, Table 14 was drawn with a more specific second approach and with the intention of improving the previous results.

Table 14

Substrate selection approach nº2

Reaction Parameters						Dosage %			
Substrate	Water	T(°C)	Regime	Time uts	pH	Cellulase	Pectinase	Amylase	Xylanase
1 Celeriac peeled roasted	Recover+20%	50	*Sous-vide*	120 min	5.5-6.4	0.2	0.2	0.1	0.1
2 Celeriac peeled roasted	Recover+10%	50	*Sous-vide*	120 min	5.5-6.4	0.2	0.2	0.1	0.1
3 JA whole boiled and roasted	Recover+10%	50	*Sous-vide*	120 min	5.5-6.4	0.2	0.2	0.1	0.1
4 Potato whole boiled and roasted	Recover+10%	50	*Sous-vide*	120 min	5.5-6.4	0.2	0.2	0.1	0.1
5 Pumpkin cored peeled roasted	Recover+10%	50	*Sous-vide*	120 min	5.5-6.4	0.2	0.2	0.1	0.1
6 Pumpkin cored unpeeled roasted	Recover+10%	50	*Sous-vide*	120 min	5.5-6.4	0.2	0.2	0.1	0.1
7 Pumpkin cored unpeeled roasted	Recover+20%	50	*Sous-vide*	120 min	5.5-6.4	0.2	0.2	0.1	0.1

Regarding the products used, they remain the same as the ones proposed in the previous test, but with a significant difference, the usage of oil when roasting. Fat had not been previously used to keep the experiments as simple as possible to identify problematics with more clarity, but as the roasting process was a critical point in the previous, a small amount of sunflower oil was added before roasting to boost Maillard reaction (2% over the weight of the product).

According to Rempel y Withers (2008), in enzymatic hydrolysis, the chemical breakdown of a substrate is strictly dictated by the number of bonds being severed, consuming exactly one water molecule for every single bond cleaved. Based on this, and as the textures observed in the previous test samples were unstandardized, the amount of water added will be as much as weight has lost during cooking plus a 10% additional. Furthermore, two samples are to be performed with a 20% additional to compare them with the rest (1 and 7).

Juice production for the fermentation

With the method complete and substrate flavors optimized, the last part of the section is specific to evaluate that the data the actual production for te vinegars. The set parameters can be seen in Table 15.

Table 15

Juice production

	Reaction Parameters					Dosage %			
Substrate	Water	T(°C)	Regime	Time uts	pH	Cellulase	Pectinase	Amylase	Xylanase
1 Boiled peeled beetroot	Recover+20%	50	*Sous-vide*	120 min	5.5-6.4	0.2	0.2	0.1	0.1
2 Roasted potato	Recover+10%	50	*Sous-vide*	120 min	5.5-6.4	0.2	0.2	0.1	0.1
3 Steamed pumpkin	Recover+0%	50	*Sous-vide*	120 min	5.5-6.4	0.2	0.2	0.1	0.1
4 Roasted celeriac	Recover+20%	50	*Sous-vide*	121 min	5.5-6.5	0.2	0.2	0.1	0.1
5 Roasted JA	Recover+20%	50	*Sous-vide*	120 min	5.5-6.4	0.2	0.2	0.1	0.1

The five chosen substrates were enzymatically processed according to the optimal parameters improved, analysed and optimized throughout the development. With only variability in the additional water, using: 20% for beetroot, celeriac and jerusalem artichoke to maximize yield; 10% for potato, to preserve flavor intensity; 0% for pumpkin, as it was decided to steam the product and it was expected to keep hydration in the product.

Regarding cooking methods: beetroots were simmered (90-95ºC) for 60 minutes, then peeled and roughly chopped; potatoes simmered for 45 minutes, chopped into big pieces, and roasted at 195ºC for 35 minutes with 2% sunflower oil; pumpkin, cored, cut into big pieces and steamed at a 120ºC for 25 minutes; celeriac, roasted whole at 195ºC with 2% sunflower oil for 70 minutes, then cut into big pieces; jerusalem artichoke, roasted at 195ºC for 30 minutes with 2% sunflower oil.

2.2.4. Phase 4: Vinegar fermentation

The vinegar fermentation method performed, and the recipes built trying to combine the next considerations:

1. The method must be simple enough to scale it into a bigger production unit that meets the restaurant's demand and it's feasible within the spaces.
2. The aimed final products organoleptic profile must be coherent with Geranium culinary work frame, which means there must be balance between sourness and sweetness and the focus must be the natural flavor of the product utilized.
3. The process needs to possibly be easily transferred and extrapolated with other ingredients.

To decide and develop the final fermentation process chosen and the different specifications demanded, an initial study of feasibility was not performed. Instead, a combination of the knowledge acquired and expressed throughout the report has been used and the firsthand knowledge of the company as a worker that has been part of the restaurant for over a year.

Next, the different pinpoints of the main parameters chosen:

Alcoholic fermentation will not be utilized. This is to attend points 1, 2 and 3: first, to keep the method simpler, shorter, less space and labor demanding; second, for the natural sugar content of the ingredient to not be transformed into ethanol and so preserve the natural flavor profile; third, for the fermentable sugars content to not be a critical point for choosing which ingredients to use.

It will be intended to have a low final amount of acetic acid formed. This is to attend point 2. Less sourness means more balance and natural flavor; shorter fermentation time means fewer secondary flavors developing but keeping enough acidity so the final product is shelf stable.

Submerged oxygenation will be utilized. This is to attend point 1 and 2. On one hand, to shorten the fermentation time. On the other hand, so the product does not develop too much complexity.

The natural temperature in the room where performed will be utilized. This is to attend point 1. For the process to remain simpler, will be performed at room temperature without adding any component to fix a specific temperature level, even if that means performance is suboptimal or fermentation time grows.

With these points exposed, this is the final description of the final vinegar developing process:

For each of the 5 chosen liquified vegetables, a 2,5L glass jar 1 kg of base liquid with an added 18% apple cider vinegar with 5% acetic acid content, and 5% alcohol over the whole weight of the mix. This fermented at an average room temperature of 21ºC with a 4,4L/min air pump regulated at minimum speed and divided for 2 jars dissolving oxygen with a porous air stone and an initial pH of minimum 4,5. This to let ferment for as long as all the alcohol is consumed, and so, the pH levels stops dropping. Tasted and checked daily to build data on the evolution until endpoint. Finally, pasteurize at a temperature of 75ºC in a water bath for 3 minutes with the previously filtered vinegar into 1L capacity bottles with 950g of vinegar inside.

Next, Table 16 with the recipe utilized.

Table 16

Vinegar recipes

		18%	5,00%		
	Liquid/juice	Mother vinegar	Alcohol 96%	Total vinegar	pH
Boiled peeled beetroot	1000	180	59	1239	3.9
Roasted potato with skin	850	153	50	1053	4.1
Steamed pumpkin with skin	900	162	53	1115	4.0
Roasted celeriac with skin	1000	180	59	1239	3.9
Roasted JA with skin	1000	180	59	1239	4.2

Note: the pH column refers to the value of the final mix.

2.2.5. Phase 5: Viability evaluation

The viability evaluation focuses on the enzymatic method rather than the vinegars. It goes through 3 sections and compares how the application of enzymes within the company's daily operations could improve existing processes. This is done by calculation with specific data of economic potential saving and waste management improvement.

It starts with the study of the juicing mechanisms already existing in the restaurant and it compares the results obtained with the ones obtained throughout the different experiments performed in the optimization of the enzymatic method.

Next, all this data is extrapolated to amount of food that could be saved weekly and yearly, not only in the production of the menu, but of the non-alcoholic pairing.

Finally, knowing all the food waste that could be saved is evaluated the economic impact that this could mean weekly and yearly. There is also a study of the producing cost for the vinegar with the entire breakdown of the cost of each part of different elements required and an analysis of how much the production cost may be improved.

3. RESULTS

The following section provides the results obtained throughout the progression of the project. The subdivisions follow the same structure as Development section, and it attends to the experiments carried out providing the objective conclusion of each part substantiated in analyzable data.

3.1. Phase 1 results

3.1.1. Enzymatic reaction time evaluation

The data obtained in the experiment is summarized in Table 17.

Table 17

Enzymatic reaction time evaluation results

Reaction Parameters						Dosage %			Results				
	Substrate	T(°C)	Regime	Time uts	pH	Cellulase	Pectinase	Amylase	Pommace(g)	Liquid(g)	Waste(g)	Yield %	Deviation
1	Celeriac pomace +100% water	50	*Sous-vide*	150 min	5.5	0	0	0	193	197	10	**-2%**	3%
2	Celeriac pomace +100% water	50	*Sous-vide*	30 min	5.5	0.2	0.4	0.5	120	261	19	**31%**	5%
3	Celeriac pomace +100% water	50	*Sous-vide*	60 min	5.5	0.2	0.4	0.5	51	320	29	**60%**	7%
4	Celeriac pomace +100% water	50	*Sous-vide*	90 min	5.5	0.2	0.4	0.5	47	331	22	**66%**	6%
5	Celeriac pomace +100% water	50	*Sous-vide*	120 min	5.5	0.2	0.4	0.5	42	346	12	**73%**	3%
6	Celeriac pomace +100% water	50	*Sous-vide*	150 min	5.5	0.2	0.4	0.5	35	342	23	**71%**	6%
											Average deviation		5%

Note: the yield % is calculated over the initial weight of pomace, not the whole weight of the mix. All percentages besides deviation are calculated based on (w/w).

Regarding the data obtained from sample 1, when the mixture was subjected to the same conditions but without the presence of enzymes, neither juice extraction nor structural degradation occurred, the obtained yield is -2% as it probably absorbed a part of the water.

When analyzing samples 2 to 6, the reaction is highly efficient within the first 60 minutes and reaches its endpoint at 120 minutes. The yield increased by 7% between minute 90 and 120 but remained constant between 120 and 150 minutes. Based on Equation 1, the reaction should have found completion after 150 minutes, but it proceeded faster than expected. This may have probably been because vacuuming improves impregnation of the enzymes that are diluted in the medium into the vegetable's pomace (Radziejewska-Kubzdela et al., 2014).

This was further evaluated with the usage of polynomial regression, with which the curve shown on Figure 8 was obtained using the yield value of the different samples of the experiment.

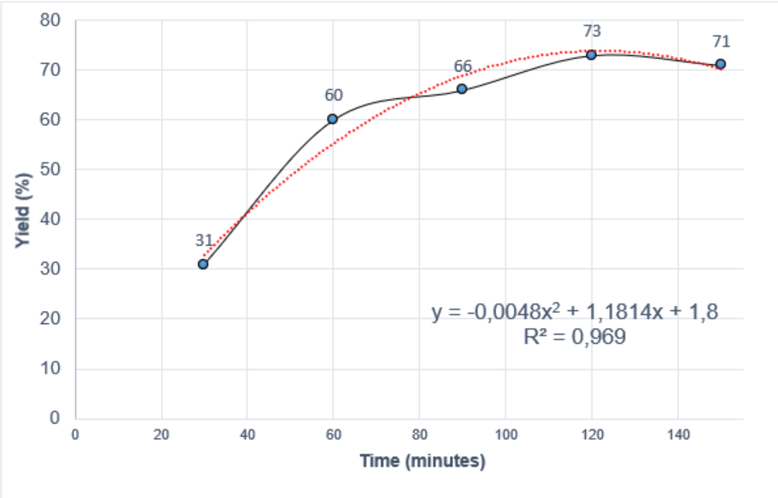

Figure 8

Enzymatic reaction evolution throughout time

It can be seen how the reaction progress is highly dependent on time, having most of the growth over the first stage of the reaction, between minute 0 and 60. From minute 60 to 120 the curve keeps growing reaching reaction completion. The curve shown follows a very similar logic to Michaelis-Menten (Srinivasan, 2022) model curve presented in Figure 4, what validates that the reaction followed the expectable performance.

Sample 5 provided the highest yield with 73%, very close to the theoretical maximum yield (81.45%), so 120 min was set as standard for the procedure. The remaining unextracted water content may be attributed to several factors: experimental errors during manual handling, deviations from optimal reaction conditions, or the presence of other liquid-retaining compounds, such as protein and hemicellulose.

3.1.2. Enzyme activity testing

The results obtained in the second of the experiments can be seen in Table 18.

Table 18

Enzyme activity testing results

Reaction Parameters			Dosage %			Results				
	Substrate	Time uts	Cellulase	Pectinase	Amylase	Pommace(g)	Liquid(g)	Waste(g)	Yield %	Deviation
1	Celeriac pomace +100% water	120 min	0	0.4	0	188	200	12	**0%**	3%
2	Celeriac pomace +100% water	120 min	0.2	0	0	182	201	17	**1%**	4%
3	Celeriac pomace +100% water	120 min	0	0	0.5	91	279	30	**40%**	8%
4	Celeriac pomace +100% water	120 min	0.2	0.4	0	158	230	12	**15%**	3%
5	Celeriac pomace +100% water	120 min	0	0.4	0.5	67	326	7	**63%**	2%
									Average deviation	4%

Note: All percentages besides deviation are calculated based on (w/w).

The table shows the obtained yields depending on the combination of enzymes on the same substrate with equal conditions. Samples 1-3 evaluate each enzyme individually, 4 and 5 in combination. Figure 9 summarizes and compares results.

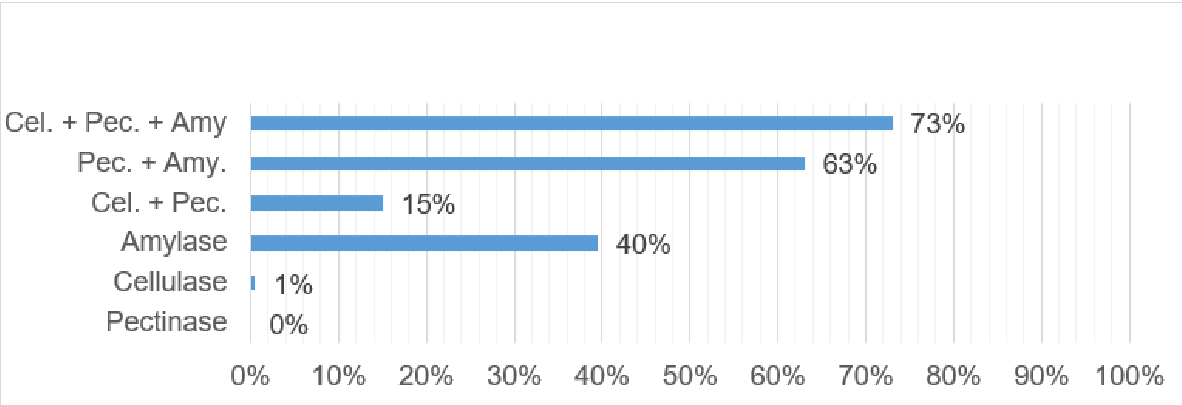

Figure 9

Enzyme activity testing yields

The results for the individually acting enzymes show that amylase produced the highest yield when used alone (40%), while pectinase and cellulase could not react with the substrate on their own (0%). However, strong synergistic effects emerged when they were combined, as pectinase and cellulase mixture achieved a 15% yield, while pectinase and amylase reached 63%.

This synergy occurs because pectinase acts on the structural "glue" of the plant tissue, degrading the middle lamella to release trapped intracellular fluids and increase matrix porosity, which allows the other enzymes to better penetrate the tissue. Ultimately, this proves that the maximum yield (73%) is achieved when all the enzymes are utilized together at optimal thermodynamic conditions: a combination of 0.2% cellulase, 0.4% pectinase, and 0.5% amylase at 50°C and a pH of 5.5 for 120 minutes.

Based on these results, amylase had the most significant effect on the process. This high efficacy is directly attributed to the starch-rich nature of celeriac (*Celeriac, Celery Root, Raw - Nutrition Data & Facts, Nutritional Information*, 2019); the rapid enzymatic hydrolysis of these starches causes an immediate structural collapse of the pomace.

Conversely, cellulase had the lowest overall impact (10%) due to the naturally crystalline and recalcitrant structure of plant cellulose, which strongly resists degradation within the 120-minute window before the reaction naturally plateaus (Zoghlami & Paës, 2019).

Because amylase is the primary driver of this liquefaction, future reactions could potentially be conducted at a higher pH to favor its specific activity. By utilizing the natural pH of celery root pomace, this adjustment could eliminate the need for acidifiers like ascorbic acid, thereby yielding a liquid with a purer flavor profile without compromising the overall efficiency of the process.

3.1.3. Liquid-solid interaction and pH

In Table 19 can be seen the results of the third experiment.

Table 19

Liquid-solid interaction results

Reaction Parameters			Results				
	Substrate	pH	Pommace(g)	Liquid(g)	Waste(g)	Yield %	Deviation
1	Celeriac pomace	6.2	67	126	7	**63%**	4%
2	Celeriac pomace + 30% juice	6.2	63	190	7	**65%**	3%
3	Celeriac pomace + 60% juice	6.2	46	263	11	**72%**	3%
4	Celeriac pomace + 100% juice	6.2	49	343	8	**72%**	2%
5	Celeriac pomace + 30% water	6.2	46	211	3	**76%**	1%
					Average yield	69%	
					Average deviation		3%

Note: All percentages besides deviation are calculated based on (w/w).

Next, Figure 10 shows a graphic to compare the yields of the different samples

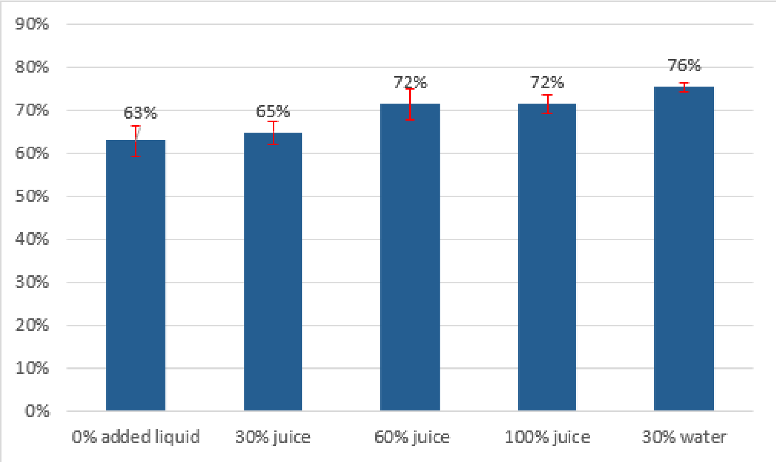

Figure 10

Yield comparation for liquid addition

Note: the (red) error marks show how the value may change (+/-) due to standard deviation.

Respectively to the yields obtained from the different samples evaluated, number 5 provided the highest with 76%. This is 11% above sample 2, which had the same amount of added liquid but used celeriac juice instead of water.

The main explanation could be that the addition of water increases the amount of calcium in the mix which provides amylase better operating conditions, however, according to Poulsen et al (2026). Danish celery root has a natural calcium content of 37mg per 100g (370ppm), already way above the 100ppm amylase requires (Bush et al., 1989), what means that calcium content cannot be the differentiator.

Another possibility would be the pH difference when adding water as it may have increased it, but this should not be a reason for yield improvement, but decrease, as the enzymes work optimally at a pH below 6.2 (X. Wang et al., 2023).

The last possibility is that, as more free water molecules are available, the enzymes have higher capacity to hydrolyze the substrate, reaching an overall higher final yield (McDonald et al., 2009). Idea that confirms when comparing that sample 1 (with no added liquid) provided 63% yield and sample 5 (with 30% added water) 76%. However, adding liquid dilutes overall flavor, and de-aligns with Geraniums culinary framework so, for upcoming experimentation it will be used a 10% added water, where to mix the enzymes.

With the reevaluation of pH, the average final yield obtained from sample 4 was 72%. This differs by 1% from the results of the first testing, where the sample had the exact same conditions, but with 5.5pH. Such a small difference suggests that using the natural pH of the product instead of setting it at 5.5 by adding ascorbic acid is not an important differentiator. For further experiments it was decided to use the natural pH of the substrate.

3.1.4. First directional tasting

To assess the preliminary viability of the prototypes, an informal directional tasting was conducted with the samples of the previous experiment. The goal was not to perform a standardized sensory analysis, but rather to gather qualitative, practical feedback to guide the iterative development process. Samples were tasted multiple times, with a water rinse between each tasting, using plain, fresh celeriac juice as a reference.

The tasting revealed that the juice obtained from the celeriac pomace was significantly more sour and slightly bitter, effect caused by the cellular walls rupture. Because the enzymatic breakdown of starches should theoretically increase the sweetness of the final liquid, it is hypothesized that the observed sourness is due to the prior mechanical extraction, which removed a large portion of the products simple sugars that are diluted in the natural water content. This hypothesis aligns with the fact that fresh, mechanically extracted celeriac juice is very sweet, an observation that is unexpected given the natural sugar content per 100g of the fresh celery root (1.71g/100g) (*Celeriac, Celery Root, Raw - Nutrition Data & Facts, Nutritional Information*, 2019).

Added on, this significant increased sourness may be due to the use of 0,25% ascorbic acid, that, besides remaining a low dosage, the presence of it will always affect flavor.

As main conclusions of the tasting: when working with previously processed vegetable trimmings, it is essential to consider how prior treatments may impact the flavor profile of the final product, and the addition of an acid to fix pH will always affect flavor. So, for upcoming experimentation it was decided to use only non-pretreated products and quit the use of ascorbic acid.

3.1.5. Refrigeration temperature testing

The results obtained from the experiment can be seen in Table 20.

Table 20

Refrigeration temperature results

Reaction parameters				Results				
Substrate	T(°C)	Regime	Time uts	Pommace(g)	Liquid(g)	Waste(g)	Yield %	Deviation
1 Celeriac pomace	4	Refrigerator	12 h	121	83	6	**37%**	3%
2 Celeriac pomace	4	Refrigerator	18 h	96	110	4	**50%**	2%
3 Celeriac pomace	4	Refrigerator	24 h	79	126	5	**58%**	2%
4 Celeriac pomace	4	Refrigerator	36 h	70	134	6	**62%**	3%

Note: All percentages besides deviation are calculated based on (w/w).

Regarding sample 4, it provided a yield of 62%. When comparing this with the celeriac sample with no added water from the previous experiment, that had 63% yield there is only a 1% difference. So, it can be concluded that the method can be applied at refrigeration temperatures, but with a large increase in the amount of time needed to reach endpoint. It would only be recommended to do when: is important to protect the products organoleptic profile from temperature; just dropping the elaboration in the refrigerator rather than keeping it on a specific temperature has enough value due to practical application to compensate time increase; or to protect it from bacterial activity proliferation if it is not intended to pasteurize afterwards.

An extra possibility would be to start the reaction at the optimal temperature and if there is not enough time to complete it because of any kitchen inconvenience it could be just dropped in the fridge overnight until the reaction reaches endpoint.

When analyzing the different yields provided by each of the samples it can be seen how reactions endpoint was reached at approximately hour 36th. As with the first experimental design, a curve has been built with the data from the experiment to evaluate the reactions performance throughout time, see Figure 11.

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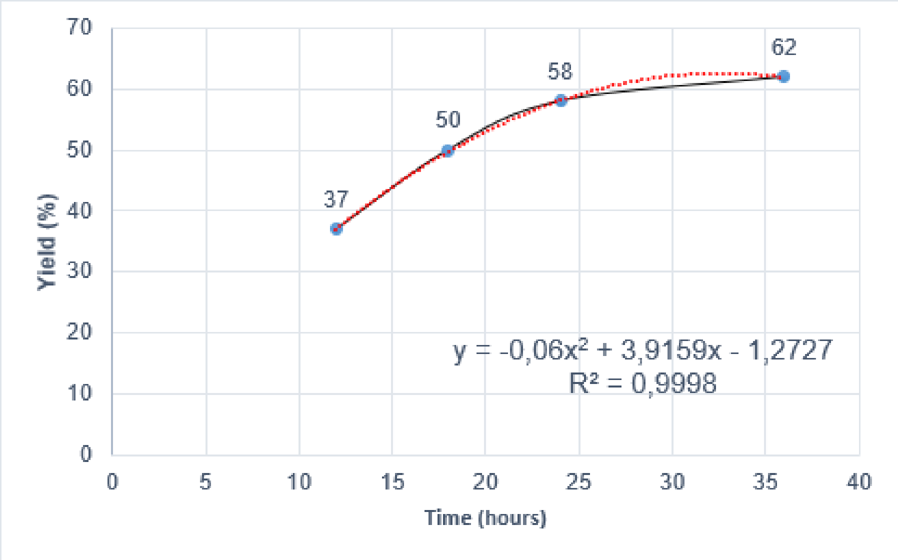

Figure 11

Enzymatic reaction evolution throughout time at refrigeration temperature

Respectively to the previous measurements, most of the reaction happens in the first half of the time, but it does not reach completion until hour 36. This means that enzyme temperature kinetics follow Michaelis-Menten curve (Srinivasan, 2022) either at optimal or suboptimal temperature.

3.2. Phase 2 results

3.2.1. Evaluation on raw generic vegetables

In Table 21, the results of the experiment are shown.

Table 21

Evaluation on raw generic vegetables results

Reaction Parameters			Results				
Substrate			Pommace(g)	Liquid(g)	Waste(g)	Yield %	Deviation
1	Carrot	Apiaceae	12	92	6	**82%**	**5%**
2	Potato	Solanaceae	33	58	5	**48%**	**5%**
3	Beetroot	Amaranthaceae	100	10	0	**0%**	**0%**
4	Cauliflower	Brassicaceae	100	10	0	**0%**	**0%**
5	Broccolini	Brassicaceae	100	10	0	**0%**	**0%**
6	Kohlrabi	Brassicaceae	100	10	0	**0%**	**0%**
7	Tarragon	Asteraceae	100	10	0	**0%**	**0%**
8	Lemon Verbena	Verbenaceae	100	10	0	**0%**	**0%**
					Average deviation		1%

Note: All percentages besides deviation are calculated based on (w/w).

The results were not as expected, with only two of the eight samples successfully enzymatically liquefied. Analyzing the reasons behind this low degradation rate revealed several specific structural barriers. The potato sample, for instance, reacted less efficiently than the carrot. Because it was completely raw, a significant portion of its starch chains were too complex for amylases to fully cleave, a limitation that could potentially be resolved by pre-cooking the vegetable.

A different challenge was observed in the cauliflower, broccolini, and kohlrabi samples, all belonging to the Brassicaceae family. These vegetables possess complex hemicellulose structures containing xylan, which block enzyme permeability and catalysis. This suggests that adding xylanase is key to facilitating the reaction in such vegetables (Fagbohoun Jean Bedel et al., 2020; Long et al., 2022).

Similarly, the beetroot sample resisted degradation due to its heavily reinforced matrix, where pectin molecules are strongly bound to calcium ions. Breaking down this highly complex structure would likely require cooking and blitzing the beetroot before introducing the enzymes (Pattarapisitporn & Noma, 2025).

Furthermore, herbs like tarragon and lemon verbena are naturally protected by a hydrophobic wax coating that renders their structures impermeable. Pre-blanching these herbs would melt the wax layer and expose the dietary fibers to the enzymes, though the absence of xylanase may also contribute to their resistance(Deng et al., 2019).

Ultimately, the structural differences across specific vegetables and botanical families demonstrate that the current method is not versatile enough to operate universally. To increase the range of products that can be liquefied raw, the addition of xylanase appears imperative, while other products will fundamentally require thermal or mechanical pre-treatment to become susceptible to enzymes. Before testing further methodology, it is necessary to evaluate other botanical families to verify which structures are inherently resistant to the current process.

3.2.2. Vegetables by botanic family

The results obtained from the experiment can be seen on Table 22.

Table 22

Vegetables by botanic family results

Reaction Parameters		Results				
Substrate		Pommace(g)	Liquid(g)	Waste(g)	Yield %	Deviation
1 Leek	Amaryllidaceae	33	74	3	**64%**	**3%**
2 Fennel	Apiaceae	45	64	1	**54%**	**1%**
3 Red bell pepper	Solanaceae	14	89	7	**79%**	**6%**
4 Zucchini	Cucurbitaceae	22	87	2	**77%**	**1%**
5 Cucumber	Cucurbitaceae	14	94	2	**84%**	**2%**
6 Pumpkin	Cucurbitaceae	10	95	6	**85%**	**5%**
7 Jerusalem artichoke	Asteraceae	47	57	6	**47%**	**5%**
				Average deviation		3%

Note: All percentages besides deviation are calculated based on (w/w).

Based on the obtained results, in which all samples produced a significant yield, the hypothesis that the method's failure on the vegetables from the previous testing was due to the structural composition of Brassicaceae appears to be true. However, it should be noted that more fibrous vegetables, such as leek, fennel and jerusalem artichoke, produced smaller yields than the other samples, reaffirming that the implementation of xylanase may improve the overall results of the enzymatic reaction.

Next, Figure 12 is a graphic with a comparison of the botanical families tested so far.

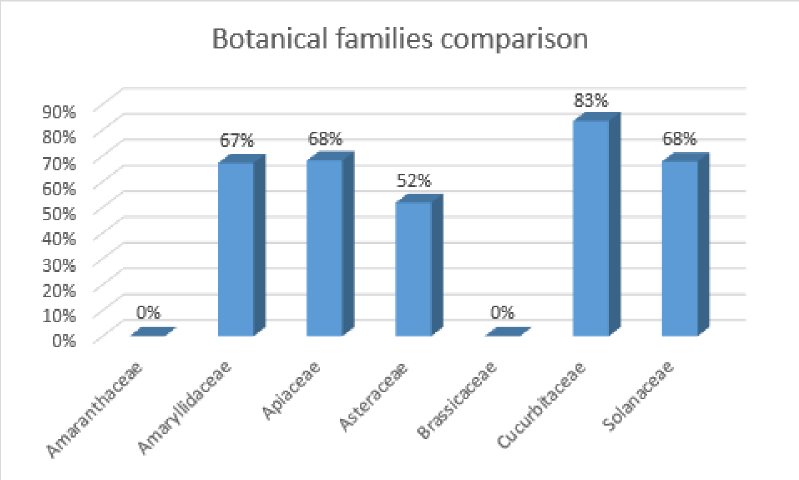

Figure 12

Botanical families comparison

The bars graphic compares the yields obtained from the vegetables of each of the different botanical families tested. To obtain the percentage of those of which multiple products were tested (such as Apiaceae with celeriac, carrot and fennel) it was used average.

Regarding the results obtained: Cucurbitaceae averaged the highest yield with 82%; followed by Amaryllidaceae, Apiaceae and Solanaceae with 64%, 67% and 66% respectively; Asteraceae with 57%; and Amaranthaceae and Brassicaceae with 0%.

Overall seems like the more fibrous the vegetable naturally is, the lower the yield, which tells that mechanically processing and cooking the vegetables may be a way to improve efficiency, as they are ways to break down the fiber structures.

Regarding Amaranthaceae and Brassicaceae, they are families to test by separate and confirm the previously mentioned hypothesis of why the enzymes may not be acting on them.

3.2.3. Second directional tasting

As previously, an informal directional tasting was conducted to obtain a general assessment of the flavor of the resulting liquids. All samples from the previous two experiments were tasted alongside Søren Ledet, co-proprietor of the restaurant and Advanced Sommelier and Ronni Vexøe Mortensen, head chef and thesis director. The main conclusions were as follows: the overall flavor of the liquids remained similar to that of the fresh products from which they were derived; allium notes in the leek were noticeably milder than they would be in the raw product or the raw, mechanically extracted juice; and cucumber sample exhibited phenolic notes reminiscent of tea fermentation. This could be a secondary effect caused by the fact that the utilized pectinase contains a part of remaining glycosidases (*Enzyme Products for Industry, Developers, Researchers and Education*, 2026).

The main feedback was that the method showed a resulting juice with a very different taste than what a mechanically extracted juice would be, as sweetness remained a lot more like the origin product, what makes the overall flavor closer to the base ingredient.

3.2.4. Vegetables when cooked and mechanically processed

The results of the experiment can be seen in Table 23.

Table 23

Vegetables when cooked and mechanically processed results

Reaction Parameters	Results				
Substrate	Pommace(g)	Liquid(g)	Waste(g)	Yield %	Deviation
1 Raw chopped jerusalem artichoke	47	57	6	**47%**	**5%**
2 Boiled chopped jerusalem artichoke	48	55	8	**45%**	**7%**
3 Boiled jerusalem artichoke purée	25	78	8	**68%**	**7%**
4 Raw chopped potato	33	58	5	**48%**	**5%**
5 Boiled chopped potato	14	83	13	**73%**	**12%**
6 Boiled potato purée	5	98	7	**88%**	**6%**
				Average deviation	7%

Note: sample 1 and 4 results, that attend the raw chopped product, were copied from previous experiments. All percentages besides deviation are calculated based on (w/w).

Regarding jerusalem artichoke, raw and cooked provided the same yield. But when processed into a purée, it increased by 21%, from 47% to 68%. Confirming the hypothesis that blitzing breaks down the structure and makes the substrate more available for the enzymes to react with it.

When comparing the potato samples 5 and 6, the same effect when blitzing can be observed as it grows 15%, from 73% to 88%. Differently from the jerusalem artichoke, cooking the potato does affect the yield, increasing it by 25%, from 48% to 73%, what confirms that cooking is a way to break down starch chains into less complex structures for the amylase to process them.

The overall conclusion is that, blitzing and cooking will make the liquefying process easier for the enzymes, maximizing final yield.

3.2.5. Brassicaceae family

The results of the experiment can be seen on Table 24.

Table 24

Brassicaceae family results

Reaction Parameters	Dosage %				Results			
Substrate	Cellulase	Pectinase	Amylase	Xylanase	Pommace(g)	Liquid(g)	Waste(g)	Yield %
1 Raw chopped broccolini	0.2	0.4	0.5	0.1	90	13	7	**3%**
2 Cooked chopped broccolini	0.2	0.4	0.5	0.1	58	50	2	**40%**
3 Raw chopped kohlrabi	0.2	0.4	0.5	0.1	81	27	2	**17%**
4 Cooked chopped kohlrabi	0.2	0.4	0.5	0.1	36	80	-6	**70%**

Note: sample 4 waste(g) value is negative as the ingredient likely absorbed water when cooking. All percentages besides deviation are calculated based on (w/w).

Comparison of samples 1 and 3 (raw chopped) to the previous control group, which yielded 0%, revealed a marked increase in extraction efficiency, reaching 3% and 17% respectively. These results support the hypothesis that xylanase facilitates the breakdown of the plant cell wall matrix by hydrolyzing hemicellulose, thereby increasing the release of the desired product (Polizeli et al., 2005). Given this efficacy, xylanase was selected for further optimization in subsequent experimentation.

The cooked samples demonstrated a significant increase in extraction yield, reaching 40% for broccolini and 70% for kohlrabi. This improvement confirms that thermal treatment effectively deactivated endogenous myrosinase, preventing the unintended hydrolysis of glucosinolates and enabling the exogenous enzymes to operate without interference (Barba et al., 2016).

The observed discrepancy in yield between the two ingredients (40% vs. 70%) likely correlates with the respective cooking durations of 3 and 14 minutes, underscoring that the intensity of thermal processing, and the resulting degree of enzyme inactivation, is a critical parameter when optimizing extraction protocols for the *Brassicaceae* family (Oloyede et al., 2021).

3.2.6. Amaranthaceae family

The results of the experiment can be seen in Table 25.

Table 25

Amaranthaceae family results

Reaction Parameters	Dosage %				Results			
Substrate	Cellulase	Pectinase	Amylase	Xylanase	Pommace(g)	Liquid(g)	Waste(g)	Yield %
1 Raw chopped beetroot	0.2	0.4	0.5	0.1	100	10	0	**0%**
2 Boiled chopped beetroot	0.2	0.4	0.5	0.1	60	50	0	**40%**
3 Raw beetroot purée	0.2	0.4	0.5	0.1	72	37	1	**27%**
4 Cooked beetroot purée	0.2	0.4	0.5	0.1	32	70	8	**60%**

Note: sample 1 results were obtained from previous experiments. All percentages besides deviation are calculated based on (w/w).

When applying the method to raw chopped beetroot the yield is 0%. This percentage grows by 40% when boiling the beetroot (sample 1 to 2), which confirms how cooking degrades the calcium-pectin structure of the vegetable and allows the enzyme to catalyze the substrate (Waldron et al., 2003). Additionally, when processing until a purée rather than chopping by knife, the yield increases by 27% (sample 1 to 3), a reconfirmation of blitzing as a mechanism to improve efficiency. The combination of both processing methods (sample 4) provided a final yield of 60%.

When observing the texture of the processed purées, it seemed like the texture was not smooth enough, and so it is hypothesized that increasing added water and cooking the beetroot for longer time (enough for the purée to come out smooth) could still significantly improve the yield.

3.2.7. Herbs

The results of the experiment can be seen in Table 26.

Table 26

Herbs results

Reaction Parameters	Dosage %				Results			
Substrate	Cellulase	Pectinase	Amylase	Xylanase	Pommace(g)	Liquid(g)	Waste(g)	Yield %
1 Raw chopped tarragon	0.2	0.4	0.5		100	10	0	**0%**
2 Raw chopped tarragon + Xyl	0.2	0.4	0.5	0.1	100	10	0	**0%**
3 Blanched chopped tarragon	0.2	0.4	0.5	0.1	90	20	0	**10%**

Note: All percentages besides deviation are calculated based on (w/w).

Comparing the results obtained it can be seen how neither of the approaches: using xylanase enzyme and cooking allowed the herbs to be liquified by the enzymes. Even though sample 3 provided a 10% yield, slightly higher, it is probably due to water absorption when cooking rather than the enzymes operating on the substrate.

After this experiment, it was decided to discard herbs and focus the remaining development exclusively on vegetables.

3.3. Phase 3 results

3.3.1. Substrate selection approach

The results of the substrate selection experiment can be seen in Table 27.

Table 27

Substrate selection approach results

	Reaction Parameters		Results			
	Substrate	Water	Pommace(g)	Liquid(g)	Waste(g)	Yield %
1	Potato whole boiled and roasted	10%	16	93	1	**83%**
2	Potato peeled boiled and roasted	10%	21	97	-8	**87%**
3	Red bell pepper roasted	10%	10	106	-6	**96%**
4	Beets boiled peeled	10%	0	104	6	**94%**
5	Beets boiled whole	10%	44	77	-11	**67%**
6	Garlic roasted pulp	10%	40	60	10	**50%**
7	Quince cored roasted	10%	59	50	1	**40%**
8	Pumpkin cored roasted	10%	35	75	0	**65%**
9	Celeriac peeled roasted	10%	60	40	10	**30%**
10	Jerusalem Artichoke whole and roasted	10%	60	50	0	**40%**
11	Jerusalem Artichoke peeled roasted	10%	50	55	5	**45%**

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Note: All percentages besides deviation are calculated based on (w/w).

Respectively to the obtained yields of the different samples, many of them showed significantly high yield (above 90%). The rest of them showed noticeably lower yields mainly due to the cooking method, which removed a lot of the present liquid in the vegetable not leaving enough water molecules for reaction catalysis. This is meant to be solved by measuring the weight difference before and after cooking and compensating for the water loss when blitzing.

Sample 7, with roasted quince showed a much different yield not because of the thickness of the purée paste, but because it naturally has a pH of 3.4, too low for the enzymes to perform the reaction.

3.3.2. Third directional tasting

An informal directional tasting was conducted with the director of the thesis Ronni Vexøe Mortensen to value the samples by their organoleptic profiles. The next conclusions were obtained: the roasted flavor profile should be rounder and more developed; and the texture of the samples would need to be standardized so there is not as much difference.

On the other hand, five samples were chosen to further improve the flavor and develop the vinegars with:

- Boiled beetroot: this sample was liked a lot and did not require flavor improvement. An important addition is to not utilize the skin as it was providing a lot of bitterness.
- Pumpkin: this was considered for improving the roasted flavor, and if not, find another cooking method which flavor profile is closer to the taste of raw pumpkin but keeping it cooked to protect it from oxidation.
- Jerusalem artichoke: the roasting profile should be improved. Both samples with skin and without were liked, but the one with skin preferred for its complexity and sustainability approach.
- Potato: the roasting profile required to be enhanced so it’s more present in the final product and skin utilized for its flavor adding properties.
- Celeriac: roasted celeriac was chosen mainly for its flavor profile.

The experiment and tasting concluded with 5 options for the development of the upcoming vinegars with the requirement of working on flavor improvement for 4 of them. Side important measures to consider are yield improvement and texture fixing by compensating water loss through cooking.

3.3.3. Chosen substrates cooking improvement

The results of the experiment can be seen in Table 28.

Table 28

Chosen substrates cooking improvement

Reaction Parameters		Cooking				Results			
Substrate	Water	Initial(g)	Final(g)	Water	add.water	Pommace(g)	Liquid(g)	Waste(g)	Yield %
1 Celeriac peeled roasted	Recover+20%	250	132	118	50	0	104	6	**94%**
2 Celeriac peeled roasted	Recover+10%	596	351	245	59,6	0	107	3	**97%**
3 JA whole boiled and roasted	Recover+10%	514	246	268	51,4	25	78	7	**68%**
4 Potato whole boiled and roasted	Recover+10%	465	268	197	46,5	10	95	5	**85%**
5 Pumpkin cored peeled roasted	Recover+10%	381	164	217	38,1	0	105	5	**95%**
6 Pumpkin cored unpeeled roasted	Recover+10%	500	246	254	50	12	93	5	**83%**
7 Pumpkin cored unpeeled roasted	Recover+20%	246	118	128	49,2	8	100	2	**90%**

Note: the "cooking" section in the table attends the initial weight, the weight after cooking and the amount of water to add to compensate the loss and additional water (either 10 or 20%). All percentages are calculated based on (w/w).

The results obtained for each substrate were evaluated individually:

First, yield and flavor wise were very similar to the previous, but the texture between sample 1 and 2 was very different. The extra 10% of water allowed the texture to transform from a fine soft purée texture to a very liquid final product.

Second, the jerusalem artichoke sample was noticeably better regarding flavor after the roasting process improvement. Recovering the water lost in the process provided a much higher yield (from 40% to 68%) and a final liquid texture.

Third, the potato sample had organoleptically improved a lot from the previous one, the liquid compensation fixed the texture issue, and the use of oil made the roasting notes noticeably rounded with a much intense and develop flavor profile.

Fourth, pumpkin, the sample with 20% added water showed a liquid texture and a final yield of 90%. The flavor was noticeably better than the one previously peeled and texture wise it was a lot better than the one with 10% added water.

With this the method to prepare the 5 substrates to produce the vinegars was set.

3.3.4. Diagram for the general application of the optimized method

The diagram next (Figure 13) shows the final general procedure for the appliance of the enzymatic reaction after all the knowledge acquired through the study.

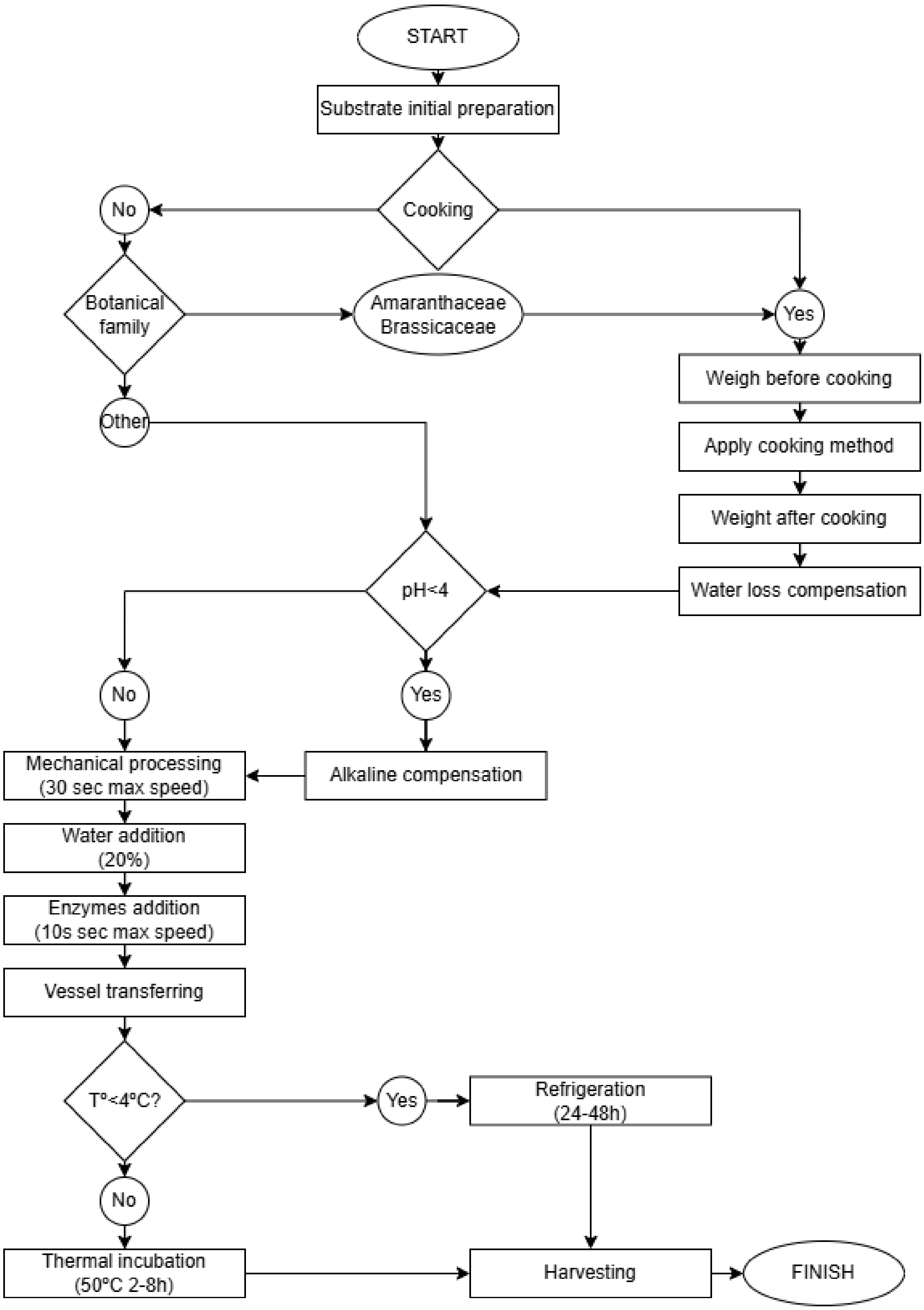

Figure 13

Enzymatic reaction diagram

The first stage of the process is the initial preparation of the vegetables. This includes operations such as peeling, washing, rinsing, coring, trimming, and brushing. And it is followed right after by the question: is cooking desired?

If the answer is yes: the cooking method must be applied to the product, very importantly, to scale the total weight before and after cooking and take note of the difference. It is important to consider the addition of other products in the cooking process that may affect the enzymatic reaction or the final liquid respectively to its final purpose (for example: fat and low acidity products).

If the product is meant to remain raw, next question must be asked: which is the botanical family? If it is part of Amaranthaceae or Brassicaceae, such as beetroot or cauliflower, it must be cooked for the liquefaction method to work afterwards.

In case of any other botanical family, then the pH level must be considered or measured. Whether cooked or raw, if the pH is below 4 the reaction will be unsuccessful or very inefficient, so it will require the addition of an alkaline aid in the mechanical processing.

At this point the substrate is ready for mechanical processing, but to do so it requires the addition of 20% water (or other liquids), and if cooked, also the amount of weight lost in the cooking process. Additionally, respectively to the weight of the whole batch 0.2% cellulase, 0.2% pectinase, 0.1% amylase and 0.1% xylanase.

The optimal way for processing is in a Thermomix or equivalent machinery for around 30-60 seconds at maximum speed, or until the particles size is small but not necessarily with a smooth texture.

With the mix ready a vessel for the reaction process must be chosen. Anything that will hold the paste/purée together should work, however, depending on the capacity for the enzymes to move through the liquid the time required until completion may increase. The best possible vessels are vacuum bags, as they improve enzyme permeability and thermal transference surface contact.

Regarding the reaction thermal incubation, it requires 2 to 8 hours at 50ºC. If desired to do at refrigeration temperature, the time required would increase to 36 to 48 hours.

At this point, the method is finished, the only thing missing would be harvesting the final product and applying any other technique required for the upcoming use, such as filtering, pasteurizing, seasoning, etc.

3.3.5. Juice production for the fermentation

The data obtained from the production to do the vinegars can be seen in Table 29.

Table 29

Juice production results

Reaction Parameters		Cooking					Results					
Substrate	Water	Initial(g)	Final(g)	Water	water+		Pommace(g)	Liquid(g)	Waste(g)	Yield %	pH	Brix º
1 Boiled peeled beetroot	Recover+20%	913	893	20	182,6	20%	0	1059	37	**95.99%**	5.8	9.38
2 Roasted potato	Recover+10%	940	566	374	94	10%	75	923	36	**88.19%**	6.1	9.47
3 Steamed pumpkin	Recover+0%	1088	954	134	0	0%	207	873	8	**80.23%**	5.5	11.44
4 Roasted celeriac	Recover+20%	1093	989	104	218,6	20%	146	1087	79	**79.45%**	5,0	8.48
5 Roasted JA	Recover+20%	1055	489	566	211	20%	185	1036	45	**78.19%**	6.3	15.36

Note: All percentages are calculated based on (w/w).

The table is of use as starting point and reference for further analysis and conclusions.

The highest yield was provided by beetroot (95.99%), but mainly as it was the only previously peeled. Potato provided 88.19%, the highest so far for this product. Pumpkin 80.24%, with a 10% loss respectively to the sample with 20% added water previously tested. Celeriac 79.45%, which means a loss of 18% respectively to previous samples with the same amount of added water, but it is not significant as those had been previously peeled. Finally, jerusalem artichoke 78.20%, 10% higher than the previous (that had 10% less added water).

It is to mention that when thieving the liquids, the only part left in the chinois was the pieces of skin/peel of the vegetables. If conversed the 36.6g of waste in the beetroot sample (which is the part that was lost during the process and vessel swapping) into percentage respectively to the initial weight of product is 4.01%, exactly the missing portion for the yield to reach 100%. So, this means that, in optimal conditions, the method would liquify the entirety of the pulp, and the higher the batch, the closer the yield will be to 100% because the part lost during the processing will be lower respective to the entirety of the weight.

3.4. Phase 4 results

3.4.1. Brix, time and pH following of the fermentation process

All the data collected in the follow-up of the fermentation process can be seen in Table 30.

Table 30

Acetic fermentation performance data

	Beetroot			Pumpkin			Celeriac			Potato			J. Artichoke		
	pH	Brixº	T(Cº)	pH	Brixº	T(Cº)	pH	Brixº	T(Cº)	pH	Brixº	T(Cº)	pH	Brixº	T(Cº)
Day 0	3.9			4,0			3.9			4.1			4.2		
Day 1	3.9	6.29	19.3	3.9	6.55	19.5	3.9	7.1	19.4	4.1	10.97	19.4	4.2	13.7	19.4
Day 2	3.9	9.36	20.1	3.9	8.64	19.7	3.9	7.34	19.7	4.1	12.65	19.7	4.2	14.22	19.7
Day 3	3.9	9.23	20,0	3.8	9.13	20.4	3.8	7.95	19.3	4.1	12.58	20,0	4.2	14.52	20,0
Day 4	3.9	9.38	19.3	3.8	9.93	19.2	3.8	8.32	19.8	4.1	12.84	19.3	4.2	14.67	19.3
Day 5	3.9	8.66	19.5	3.7	9.2	19.4	3.7	8.08	19.9	4.1	12.23	19.7	4.1	15.27	19.7
Day 6	3.9	8.73	20.07	3.7	9.27	21.5	3.7	8.01	21.6	4,0	12.04	21.6	4.1	15.17	21.8
Day 7	3.9	9.63	21.5	3.6	9.45	21.4	3.7	8.47	22.2	4,0	14.38	22.2	4,0	15.43	22.2
Day 8	3.8	9.27	21.8	3.6	9.36	21.1	3.6	8.3	21.7	3.9	14.57	21.8	4,0	15.12	21.9
Day 9	3.8	8.46	21.8	3.5	9.45	22.1	3.6	8.37	21.6	3.9	12.74	22.1	3.9	15.17	22,0
Day 10	3.8	8.54	21.1	3.5	9.5	21.4	3.6	8.56	21.05	3.9	12.98	21.25	3.9	15.32	21.2
Day 11	3.8	8.62	20.4	3.5	9.55	20.7	3.6	8.75	20.5	3.8	13.23	20.4	3.9	15.47	20.4
Day 12	3.8	8.84	19.9	3.4	9.83	21,0	3.5	8.9	20.2	3.8	13.2	21,0	3.8	17.15	21,0
Day 13	3.7	8.79	20.6	3.4	9.87	20.4	3.5	9.07	20.2	3.7	13.01	21,0	3.8	15.55	20.5
Day 14	3.7	8.89	22.4	3.4	10.14	22,0	3.5	8.97	23,0	3.7	13.07	23,0	3.7	15.68	23.4
Day 15	3.7	8.98	22.8	3.4	10.24	22.4	3.4	9.01	22.3	3.6	13.33	22,0	3.7	15.22	22.7

Note: the lines highlighted in red are those days on which the data could not be obtained because the restaurant was closed, so it was used averaging of the data from the previous day and the one after to obtain the number. This so the graphics shown next do not have blank spaces.

Based on the previous table, three graphics were built to analyze each of the data and compare vinegars. The first, regarding pH, can be seen on Figure 14.

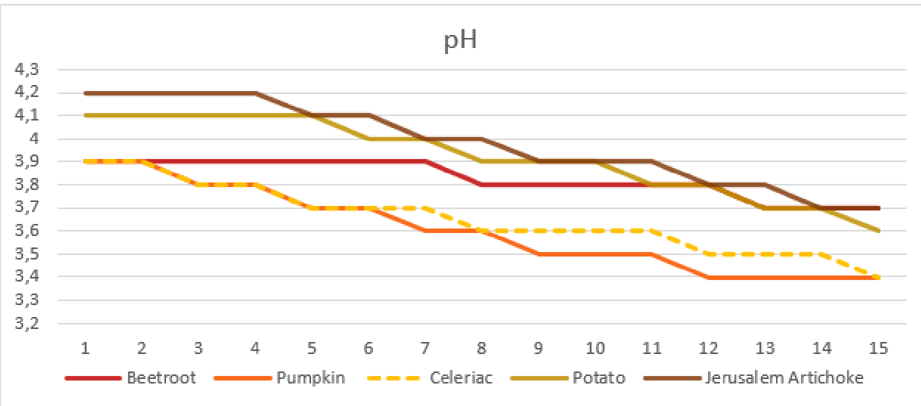

Figure 14

pH evolution

The graphic shows how pH decreased at a constant pace until day 15, when reaching reaction completion. The same measure was checked for 5 days afterwards (day 16 to 20), but the value did not change in any of them. This means that 5% of added alcohol was fully consumed and meant an alteration on the pH of -5 points for all of them, besides pumpkin (-6) and beetroot (-2).

When comparing the pace of each line, it can be seen how celeriac and pumpkin dropped the first point by day 2 meanwhile the rest took longer, what tells that the acetic bacteria, originally from apple vinegar, took longer to adapt and reproduce in those environments.

Regarding why the beetroot sample only decreased -2 in the acidity level, it likely is due to that when enzymatically breaking the structure positively charged calcium ions are freed into the environment, which neutralized the hydrogen ions that are being measured by the pH meter (Pattarapisitporn & Noma, 2025).

The second, regarding brixº can be seen on Figure 15.

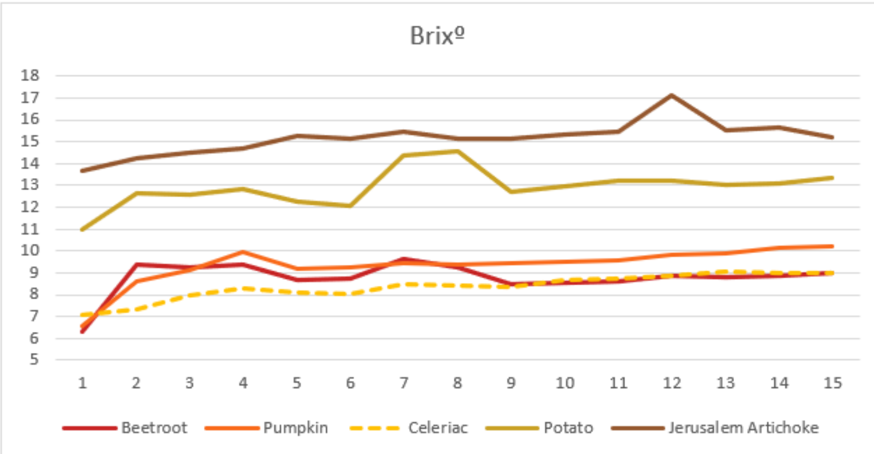

Figure 15

Brixº evolution

All the lines in the graphic that present the brixº evolution increase while the expected would be a decrease, because the acetic acid bacteria consume sugar to multiply. As the fermentation did perform, it could be explained by the fact that the (still active) enzyme keep degrading the substrate and bringing new diluted solids, which counters the effect of bacteria.

This can be clearly seen in the first 2 days, where because the 18% vinegar addition (a lot of new water molecules) the brixº significantly increased in most of the samples (Nguyen & Nguyen, 2018).

The third, regarding temperature, can be seen on Figure 16.

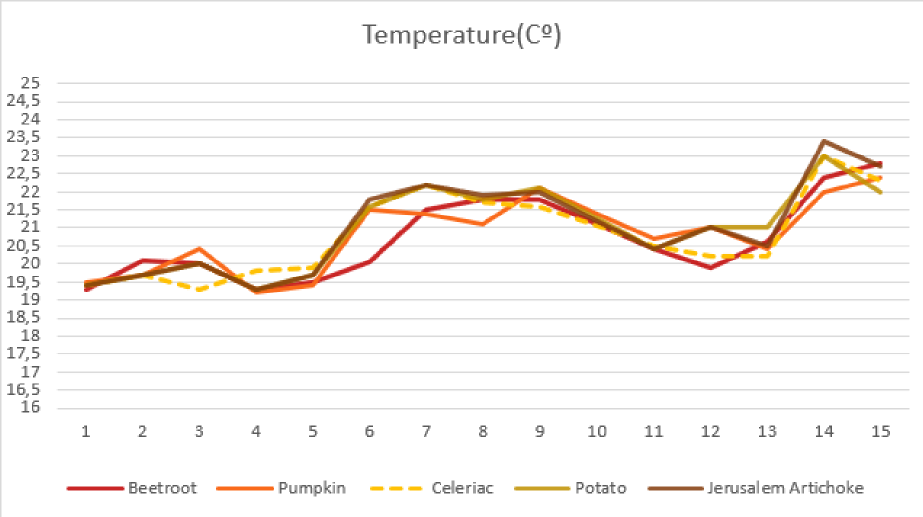

Figure 16

Temperature (Cº) evolution

The temperature level always remained at an effective temperature for the acetic fermentation. As it was performed in the month of April, it can be seen how from day first to last there was an increase of 4ºC. This is an improvement, as the optimal temperature would be between 25-30ºC. Because the restaurant's temperature does not drop below 19ºC even in winter days, the fermentation could be performed regardless of the season. However, between summer and winter, the overall needed time may be shortened by a few days.

3.4.2. Final formulated Vinegar recipes

A generic recipe for the entirety of the process since the raw vegetable to the pasteurized vinegar has been formulated with the intention of it being used as reference to be used regardless of the product. This can be seen in Annex 5 together with a recipe of boiled beetroot vinegar and baked celeriac vinegar (Figure 27, Figure 28, Figure 29).

3.5. Phase 5 results

3.5.1. Restaurant's juice yield computation

The next of the results expresses the general juice yield computation of the restaurant. It proves how the mechanical extraction of juice provides a very small yield respectively to the whole weight of the products. The data can be seen in Table 31 and reflects a week of production.

Table 31

Juicing yield computation

PRODUCT	WEIGHT(kg)	YIELD(%)	PRODUCT	WEIGHT(kg)	YIELD(%)
CELERIAC			PUMPKIN		
Whole	27,0	**100,0%**	Whole	30.5	**100,0%**
Fruit	18.1	**66.8%**	Peel	7.1	**23.2%**
Peel	8.8	**32.4%**	Fruit	19.3	**63.1%**
			Seeds	3.8	**12.5%**
Pomace	10.5	**39.1%**	Pomace	11.2	**36.7%**
Juice	7.1	**26.1%**	Juice	7.9	**25.8%**

Note: calculations have been done on (w/w).

In the juicing of celeriac 32.5% weight is lost in the peeling process, and out of the remaining fruit, only 7kg of juice are obtained, which is 26.1% of the whole product.

A similar scenario happens with pumpkin, where 35.8% of the product is lost in the peeling and coring process. And out of the remaining 7.8 liters of juice are obtained, which is 25.8% of the whole weight of the product.

Next, in Figure 17 a graphic that summarizes the data of how the yield evolved during the development process, and regarding the celeriac samples

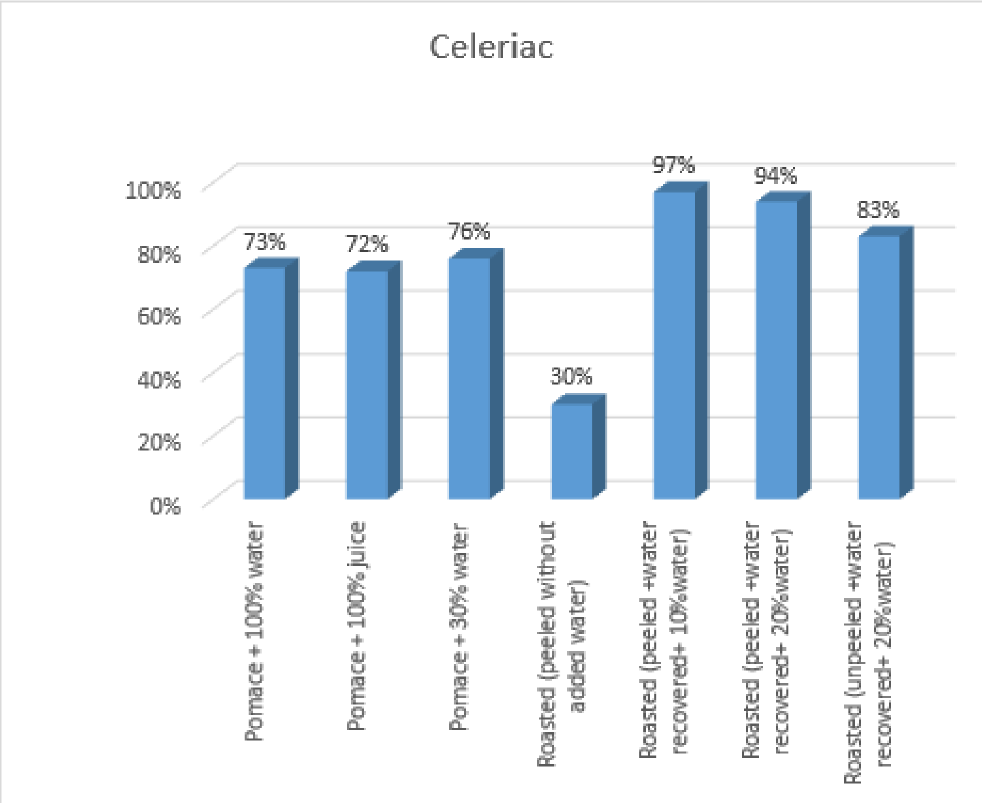

Figure 17

Celeriac enzymatic juicing yield

The different tests showed a variety of yields, but the last one, the one from the juice production for the vinegar, has 83% when the whole product was utilized. This, when compared with the yield from mechanically extracting, means an improvement of 56.9 points out of the whole product, from 26.1% to 83%.

Same exercise was performed with pumpkin and can be seen in Figure 18.

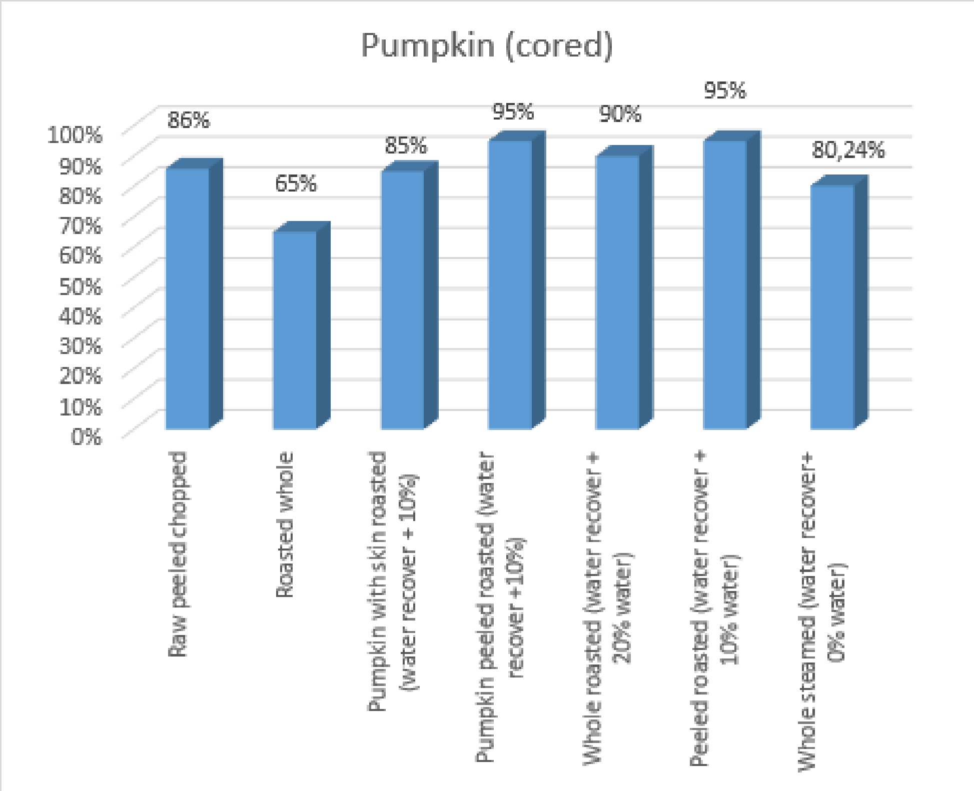

Figure 18

Pumpkin enzymatic juicing yield

When analyzing the pumpkin, the operation is slightly different, as it had been previously cored. The part of the seeds, as previously shown, attends 12.5% over the whole weight of product: because the 80.2% yield shown on the figure attends when the pumpkin is already cored (without seeds), the actual yield over the entirety of the product would be 70.2%, still far from the 25.8% of the mechanical processing. If utilizing the method that worked the best, which was with a 10% water addition (95% yield), the number would rise from 70.2% to 82.1%.

For the analysis of both cases Table 32 was utilized.

Table 32

Yield improvement analysis

PRODUCT	WEIGHT(kg)	YIELD(%)	PRODUCT	WEIGHT(kg	YIELD(%)	
CELERIAC			PUMPKIN			
Whole	27	100,0%	Whole	30.5	100,0%	
			Seeds	3.8	12.5%	
Juice (mechanical)	7.1	**26.1%**	Juice (mechanical)	7.9	**25.8%**	
Juice (enzymatic)	22.4	**83,0%**	Juice (enzymatic)	24.5	80.2%	(when cored)
			Juice (enzymatic)	21.4	**70.2%**	(when whole)
Improvement		56.9%	Improvement		54.4%	

Note: calculations have been done on (w/w).

In the table it can be seen highlighted the yield percentage of both juicing methods (mechanical and enzymatic), together with the amount of kg that would be theoretically obtained from the same amount of substrate. At the bottom the percentage respective to the yield improvement, which is the difference between the highlighted percentages.

3.5.2. Environmental potential

The analysis of the environmental potential of the method formulated in the restaurant is divided into three parts:

First, the impact that is direct and can be calculated based on the data previously collected (Table 32). If in the restaurant it is produced weekly 7.1 kg of celeriac juice and 7.9 of pumpkin, when doing the division with the yield percentages of the enzymatic juicing method (83.0% and 70.2%) it is obtained that the required amount of product is 8.5 kg of celeriac and 11.2 kg of pumpkin. This highly differs from the 27 kg of celeriac and 30.5 kg of pumpkin that are currently being utilized, saving 37.8 kg of product per week. But this is only when analyzing the celeriac and pumpkin juices, that are an individual elaboration for the sauce of two of the courses on the menu.

Second, and as continuity of the previous, what if the new juicing method was applied for the non-alcoholic pairing, that consists of mechanically juiced fruit or vegetables (either cooked or raw). There are currently 7 different juices that build the pairing, and for each about 40kg of raw product are required weekly. Assuming a conservative average yield of 60% for the enzymatic method, when doing the math as within the previous paragraph 158kg of product could be saved weekly.

Third, the impact on the menu. As mentioned, geranium is a 3 Michelin starred restaurant which's storytelling has a lot to do with deliciousness and aesthetic. To reach the standards that have been set throughout the years, thrashing vegetables trimmings is unavoidable, as food is required to look a specific and precise way. This is a common thing in restaurants this category, but because of Geranium being plant based, the part that goes to the bin is mostly vegetables. So, the enzymatic method could potentially become a standard technique for bringing back those trimmings to the dish in many different shapes, for instance, like vinegar.

3.5.3. Economic potential

The analysis of the economic potential is based in three parts: enzymatic method cost, juice making costs and vinegar costs.

The costs of the enzymatic method can be seen in Table 33.

Table 33

Enzymatic method costs

HIGH DOSAGE

Enzyme	Price per kg	Price per g	Required quantity (g/kg)	Cost	
Cellulase	1488	1.5	2	3.0	
Pectinase	1364	1.4	2	2.7	
Amylase	960	1.0	1	1.0	
Xylanase	1576	1.6	1	1.6	
			Total cost per kg of product	8.2	DKK

SMALL DOSAGE

Enzyme	Price per kg	Price per g	Required quantity (g/kg)	Cost	
Cellulase	1488	1.5	0.5	0.7	
Pectinase	1364	1.4	1	1.4	
Amylase	960	1.0	0.5	0.5	
Xylanase	1576	1.6	0.5	0.8	
			Total cost per kg of product	3.4	DKK

Note: calculations have been done on (w/w).

The table calculates the total cost of the enzymes mix based on the price per kg of each and the amount required to process a kg of substrate. The first sets a total of 8.2 DKK/kg, this is when the dosage of enzymes is high (the one utilized in the tests). The second part does the exact same but with reduced dosage (3.4DKK/kg), that is still in the recommended amounts, so it is theoretically sound but with an increase in the time required for the reaction.

Next, in Table 34 the details of how the costs for juice making have been calculated.

Table 34

Juice making cost

PRODUCT	WEIGHT(kg)	YIELD(%)	PRODUCT	WEIGHT(kg)	YIELD(%)
	CELERIAC			PUMPKIN	
Product amount	27	100,0%		30.5	100,0%
Juice (mechanical)	7.1	**26.1%**		7.9	**25.8%**
Juice (enzymatic)	22.4	**83,0%**		21.4	**70.2%**
				Cost per kg	
Whole celeriac	16 DKK		Whole pumpkin	21 DKK	
Enzymes mix	8.2 DKK			8.2 DKK	
Juice (mechanical)	61 DKK			81 DKK	
Juice (enzymatic)	29 DKK			42 DKK	
				Weekly cost	
Juice (mechanical)	432 DKK			641 DKK	
Juice (enzymatic)	205 DKK			328 DKK	
Cost difference	52%			49%	

Note: calculations have been done on (w/w).

The section "cost per kg" evaluates the cost of making a kg of juice for both the mechanical and enzymatic method. Both are calculated by multiplying the price of the substrate by the amount of it and dividing it by the quantity of juice produced. So, in the case of celeriac, it is (16 times 27 divided 7.1) for the mechanically extracted. For the enzymatically extracted the same, but with a cost added of 8.2 DKK per kg of substrate (16 times 27 plus 8.2 times 27 divided by 22.4).

When extrapolating this number to weekly costs (the cost per kg multiplied 7.1 or 7.9) it can be seen how in both cases the overall cost is halved. In the case of celeriac juice reduced by 52% and pumpkin 49%.

This in absolute terms means saving 539 DKK weekly, that becomes significant when extrapolated yearly (28.000DKK). But would be a lot more if extrapolated to halving the cost of juice making for the non-alcoholic pairing, that previously has been shown how it works with much higher amount of product.

The missing part is regarding the costs of the vinegars, and so first the cost of making the base liquid to be acetic fermented. This can be seen in Table 35.

Table 35

Vinegar base cost

Vinegar	Product C./kg	Qu.required	Added water (% w/w)		Efficiency	Qu.clean	Cost
Boiled beetroot	25	0.87	0.174	20%	95.99%	1	21.75 DKK
Roasted potato	22	1.03	0.103	10%	88.19%	1	22.66 DKK
Steamed pumpkin	21	1.25	0	0%	80.24%	1	26.25 DKK
Roasted celeriac	16	1.05	0.21	20%	79.45%	1	16.8 DKK
Roasted JA with	25	1.065	0.213	20%	78.20%	1	26.63 DKK

Note: calculations have been done on (w/w).

In the table is calculated the cost to obtain a kg of clean liquid based on the prices at which we acquire each of the products.

Next, Table 36 describes the overall cost of making the vinegar.

Table 36

Vinegar cost

Vinegar	Base C./kg	In.Qu	C.Enzymes	Alcohol C/kg	5% Qu.	Cost	M.vinegar C/kg	18% Qu.	Cost	C.total	Qtotal	C./kg
Boiled beetroot	21.75	1	8.2	330.4	0.05	16.5	128	0.18	23.04	69.5	1.23	57 DKK
Roasted potato	22.66	1	8.2	330.4	0.05	16.5	128	0.18	23.04	70.4	1.23	57 DKK
Steamed pumpkin	26.25	1	8.2	330.4	0.05	16.5	128	0.18	23.04	74.0	1.23	60 DKK
Roasted celeriac	16.8	1	8.2	330.4	0.05	16.5	128	0.18	23.04	64.5	1.23	52 DKK
Roasted JA with	26.625	1	8.2	330.4	0.05	16.5	128	0.18	23.04	74.4	1.23	60 DKK
											Average	57

Note: calculations have been done on (w/w).

With the table an average final cost of 57 DKK per kg of vinegar was obtained. The basis for the calculation is adding to the cost of the base the cost of enzymes, alcohol 96% and mother vinegar.

This value is specific to the vinegar produced in the experiment, but when applying to reality changes in the processing would lower it. The same exercise as the previous table was repeated with the beetroot sample but with different scenarios, the breaking down can be seen in Figure 19.

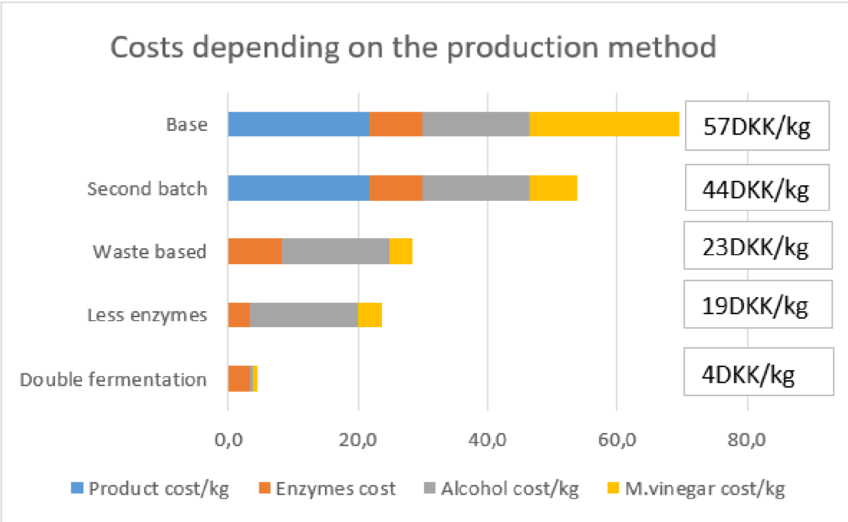

Figure 19

Beetroot vinegar costs depending on the producing method

Note: calculations have been done on (w/w).

The "base" bar divides in the costs of the one produced. The "second batch" is very similar, but with an important decrease in the cost of the mother vinegar (from 57 to 44), as instead of using store bought raw vinegar it would be used a backslope of the previous batch.

The most significant change comes from the "waste based". The previous have in consideration using bough product, but the project aims for building a method to operate on the trimmings that are meant to go to the bin, so using those would reduce the overall cost to 23 (it is considered a price of 0DKK as it is food that would anyway go do the bin).

Additionally, "less enzymes" has a final value of 19, only 4 below the previous, because of using the low dose enzyme mix. Lastly, the "double fermentation" considers all the previous but using added sugar to double ferment instead of adding alcohol.

Besides the fact that what primarily might seem like the obvious choice is the last (4DKK per kg), reducing the number of enzymes and doing alcoholic fermentation would mean a lot more labor and about double the amount of time for the process (from around two weeks to over a month). What sounds the most coherent with the project aim and the restaurant operations is using the "waste based", which means applying the same exact method studied, but using as substrate those parts of the product that would anyway go to the bin.

3.6. Gastronomic application

As an exemplification of the enzymatic liquefaction method culinary possibilities a recipe for the restaurant was developed in which the technique studied in the report is applied in the different elaborations that build the dish and pairing that can be seen in Figure 20.

Figure 20
Beetroot dessert

The dish and pairing focus on the utilization of boiled beetroot trimmings. It is built inside the ceramic as layers, from bottom to top: a walnut compote, a walnut milk foam, chopped roasted walnuts, a gel of boiled beetroot and a gastric sauce based on beetroot juice seasoned with beetroot vinegar. As pairing, beetroot and rhubarb home-made wine. The recipe can be seen in Annex 6.

In Figure 21 the different elements can be seen individually.

Figure 21
Recipe elaborations

4. CONCLUSIONS AND FUTURE STUDY LINES

The upcoming section defines all the conclusions drawn from the developing and results analysis process. These are explicitly explained divided as: objective accomplishment, learning contribution, further investigation and finished with a section of how this investigation stands out and why it is important.

4.1. Objectives accomplishment

The main objective set was reached within the stablished time. It was developed a method that allows the liquefication of most vegetables when raw or cooked that provided a result product with a very liquid texture that can be utilized for multiple culinary purposes, in this case, vinegars. Apart from Brassicaceae and Amaranthaceae botanical families, that could not be liquefied when raw.

The reaction happens in a space of 2 hours with an enzyme dosage per kg of product of 0.2% cellulase, 0.2% pectinase, 0.1% amylase and 0.1% xylanase at 50ºC and completely liquefies vegetables when adding 20% water and in the natural acidity range of the substrate unless this is below 4 ph.

This was achieved thanks to the completion of the secondary objectives:

Where the process was first optimized to define best parameters for yield efficiency and flavor.

Then it was tested on multiple substrates and conditions, and the method was reshaped for it to be reproducible regardless of the vegetable utilized.

Acetic fermentation was executed to transform the technique into a final product that can fit a space in the development of new recipes for the restaurant and that is coherent with the flavor and storytelling framework of the restaurant.

Finally, the feasibility was evaluated by comparing specific data of the amount of food waste that could be saved and the economic potential this could have.

4.2. Learning contribution

The main learning from the project is the extrapolation of a biological waste management method utilized by food industry to a fine dining restaurant context, providing a process that can be easily performed by a trained chef and that can transform both whole vegetables or trimmings into a flavor rich product with yields around 80-95% respectively to the whole weight of the product.

It was developed a generic recipe to produce vinegar out of raw or cooked vegetables juice/liquid that will work regardless of the chosen product and that will reach completion in 14 days, together with a costs sheet that can be used as estimation depending on the desired production method.

Finally, there were explored other possibilities and outsources in culinary application. Further setting the method as a culinary technique rather than an industrial process, which makes the resulting product much closer to the gastronomic field.

4.3. Further investigation

The investigation still has branches that could be further explored, next, some examples:

Flavor can be improved and standardized. In the report it is explained how the flavor remains very similar to the original of the product, but it develops slight bitterness due to the breakdown of cellular walls.

For the liquefication to perform optimally the flavor was diluted with the addition of 10-20% water. But I observed (not evaluated) how some vegetables do not necessarily require so much or can even be done with the natural water content.

Many other culinary applications can be tested. My way of seeing it is more of as a culinary technique and the possible outsources a chef can imagine are infinite, for instance, some ideas: making sauces, stews, beverages, reductions, glazes, macerations, gels, candy and many others.

3.4.3. Standing out points

Liquefying vegetables is nothing new, as it can be reached in different ways, but doing it with biologic degradation with enzymes stands out in specific points when comparing it with other technologies.

First, the yields obtained and the fact that it can be used regardless of products shape or rigidness, which is to consider when mechanically juicing.

Another would be flavor wise, the obtained liquid is not better or worse tasting than the one that you would obtain with other methods, but different, and as a chef that is an important thing to consider, as you would do when applying any other kind of cooking method.

Lastly, and which seems most differentiator, the fact that it will work after applying cooking to the product. This is a big consideration because mechanical juicing and others will not work if the product is cooked. For example, it is not possible to do celeriac vinegar from mechanical juicing, because when cooked a juicing machine will produce a thick paste, and raw celeriac juice is very susceptible to oxidation.

5. PERSONAL REVIEW OF FINAL DEGREE PROJECT

On a personal and professional level, completing this Final Degree Project has represented a defining turning point in my career, allowing me to build and exercise the capacity of bridging industrial procedures and the creative rigor of fine dining, what strengthens my capacities as a gastronomic sciences expert.

Learning to manipulate enzymatic processes to achieve yields of up to 95% was not just an exercise in resource optimization; it was a discovery of how science can amplify the identity of a raw ingredient. The fact that I developed a method that succeeds where mechanical juicing fails, particularly with cooked vegetables, has provided me with a competitive and critical edge. It taught me that true innovation often lies in adapting technologies from other sectors to solve culinary challenges.

In the professional realm, this research has deepened my commitment to circular gastronomy. Seeing how trimmings and "waste" can be transformed into vinegars with a complex flavor profile.

Ultimately, and what makes me proudest, is that I brought new knowledge and the exploration and application of culinary techniques that had never been tested in one of the most awarded restaurants in the world. What began as a liquefaction method has revealed itself to be a transversal culinary technique with infinite. I feel motivated and prepared to continue exploring this intersection of food science and gastronomy, convinced that sustainability and flavor are not just compatible, but must be the dual engine driving the kitchen of tomorrow.

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7. ANNEXES

ANEXOS / ANNEXES

INDEX OF ANNEXES

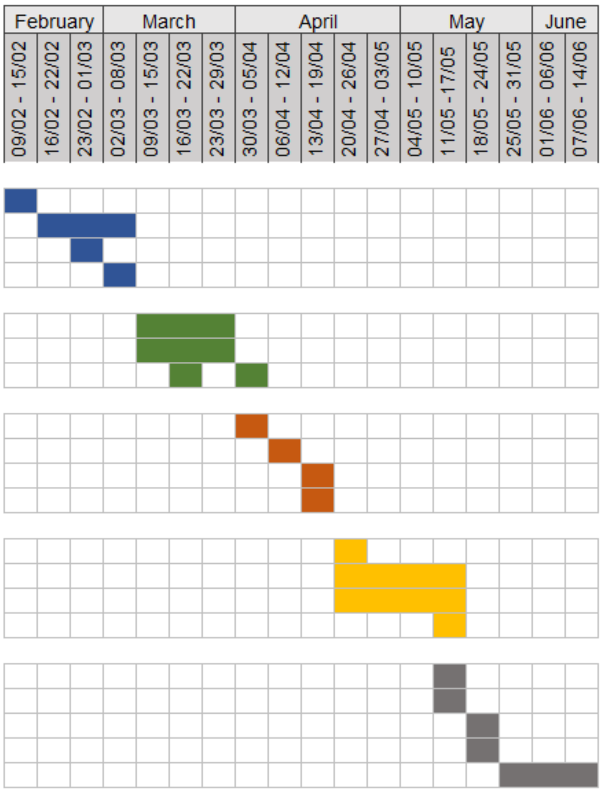
7. Anexes

Annex 1. Phases of the project

Table 37

Gantt project phases

TASK	START	FINISH
Phase 1: Research and experimental design		
Research background	9-Feb	15-Feb
Literature review	16-Feb	8-Mar
Selection of vegetable substrates	23-Feb	1-Mar
Experimental design definition	2-Mar	8-Mar
Phase 2: Enzymatic treatment evaluation		
Enzymatic reaction process optimization	9-Mar	29-Mar
Measurements of extraction yield (%)	9-Mar	29-Mar
Evaluation of the flavor potential of the samples	16-Mar	5-Apr
Phase 3: Evaluation on different substrates		
Testings on generic substrates	30-Mar	5-Apr
Process optimization for non-operative substrates	6-Apr	12-Apr
Evaluation of the flavor potential of the samples	13-Apr	19-May
Selection of the final substrate and treatment	13-Apr	19-May
Phase 4: Fermentation performance assesment		
Preparation of the final substrates	20-Apr	26-Apr
Acetic fermentation of the products	21-Apr	17-May
Process stability assesment	22-Apr	17-May
Data evaluation of the fermentative process	11-May	17-May
Phase 5: Results and feasibility		
Feasibility discussion	11-May	17-May
Potential environmental impact quantification	11-May	17-May
Determination of optimized strategy application	18-May	24-May
Final revision	18-May	24-May
Presentation and defense preparations	25-May	14-Jun



Note: the diagram divides phases and tasks along the four months duration per weeks.

Annex 2. Materials list

Table 38

Materials list

Item	Description	Figure
3,000 ml square wire clamping glass jar [Fido]	Used as vessel for the acetic fermentation	
TP603 Thermometer [ThermoPro]	Used for temperature measuring	
pH Meter for Hydroponic Water [GIDIGI]	Used for measuring the pH levels of the samples	
Digital Refractometer with ATC 0~95% Brix [LAFmate.]	Used for measuring the Brixº content of the samples	
Expressive series combi/steam oven GS451120 [Gaggenau]	Used for roasting the vegetables	
Compact Aquarium Air Pump [FEDOUR]	Used for the oxygenation of the vinegar jars	

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Mondragon Unibertsitatea — Gastronomia Zientzien Fakultatea / Facultad de Ciencias Gastronómicas / Faculty of Gastronomic Sciences — basque culinary center

Sous Vide Pro Ref. 69192 [Lacor] Ref. 69192	Used for the samples incubation	
Disinfectant [Sonett]	Used for vessels and utensils desinfectation	
1 L Demineralized Water [Borup]	Used for rinsing the measuring devices	
Apple Cider Vinegar with living Mother 5% acetic acid [Renee Voltaire]	Used as mother vinegar	
Cellulase high activity enzyme [Scientific & Technical]	Used for the bioprocessing of the vegetables	
Amylase high activity enzyme [Scientific & Technical]	Used for the bioprocessing of the vegetables	
Pectinase high activity enzyme [Scientific & Technical]	Used for the bioprocessing of the vegetables	

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Xylanase high activity enzyme [Scientific & Technical]	Used for the bioprocessing of the vegetables	
Vacuum bag 16 x 20 cm 25 pcs SOUS VIDE [Lacor]	Used for as vessel for the incubation process	
Digital pocket scale $200\times0.01g$ [Ascher]	Used for the scaling of the enzyme dosages	
Thermomix® TM5 [Vorwerk]	Used for mechanically processing the vegetables	
1000ml Glass Bottles with Swing Top [Praknu]	Used as pasteurization and conservation vessel	

Mondragon Unibertsitatea
Gastronomia Zientzien Fakultatea
Facultad de Ciencias Gastronómicas
Faculty of Gastronomic Sciences
basque culinary center

Annex 3. Enzymes technical data sheets

Amylase Technical Datasheet

1. Product Overview

Our amylase is a high temperature α-amylase produced by submerged fermentation of *Bacillus licheniformis*. The enzyme is an endo-amylase, hydrolysing 1,4-α-glycosidic linkages in starch at random, producing dextrins of different chain lengths and oligosaccharides, resulting in a rapid decrease in the viscosity of starch solutions. This enzyme hydrolysis starch into simpler sugars under high-temperature conditions commonly found in food processing. Its thermal stability allows it to retain activity during processes such as baking, brewing, and syrup production, where conventional enzymes might denature. Key benefits include enhanced starch conversion efficiency, improved product consistency, and reduced processing time and energy costs. It also supports higher yields of fermentable sugars in brewing and ethanol production and contributes to smoother textures and uniform sweetness in processed foods. Being food grade, it is safe for use as a food processing aid and compliant with regulatory standards, making it ideal for applications across the bakery, beverage, and food manufacturing industries.

2. Product Characteristics

Parameter	Specification
Enzyme Class	Hydrolase – α-Amylase (EC 3.2.1.1)
Source	*Bacillus licheniformis*.
Appearance	Brown liquid
Activity	30,000 U / mL
Units	Where 1 U is the amount of enzyme required to liquify 1 mg of soluble starch per minute
Total Protein	10-14 % w/v
Density	1.05-1.20 g per mL
Temperature Range	40–105 °C (operational)
Optimal Temperature	85–95 °C
pH Range	5.0 – 8.0
Optimal pH	5.8 – 6.5
Stability Enhancers	Ca^{2+} required for structural stabilization (typically 50–100 ppm)
Shelf Life	12–24 months (sealed, recommended storage conditions)

Figure 22

Amylase Technical Datasheet

(Source): (*Enzyme Products for Industry, Developers, Researchers and Education*, 2026).

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3. Specifications

Parameter	Specification
Enzyme Activity	>15,000 U/g
pH Range	3.5 – 6.0
Optimal pH	4.5 – 5.0
Temperature Range	30–60°C
Optimal Temperature	50–55°C
Solubility	Fully soluble in water (liquid), dispersible (powder)
Heavy Metals	< 10 ppm
Arsenic	< 1 ppm
Lead	< 2 ppm
Microbial Limits	Meets FCC / FAO / WHO specifications

Figure 23

Cellulase technical datasheet

(Source): (*Enzyme Products for Industry, Developers, Researchers and Education*, 2026).

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Gastronomia Zientzien Fakultatea
Facultad de Ciencias Gastronómicas
Faculty of Gastronomic Sciences

Pectinase Technical Datasheet

1. Product Overview

A high-activity food grade pectinase enzyme is a versatile biocatalyst widely used in the food and beverage industry for its ability to break down pectin, a complex polysaccharide found in plant cell walls. Its primary applications include fruit juice clarification, where it improves juice yield and clarity by reducing viscosity and preventing haze formation; wine production, enhancing extraction of colour, aroma, and flavour while facilitating filtration; and fruit-based product processing, such as jams, purees, and nectars, by softening fruit tissue and improving texture. Key benefits of this enzyme include increased process efficiency, higher product yield, improved sensory qualities, and reduced energy consumption during processing. Additionally, as a food-grade enzyme, it is safe for consumption and supports cleaner, more sustainable production methods by reducing the need for chemical clarifying agents.

2. Product Characteristics

Parameter	Specification
Enzyme Class	Hydrolase – (EC 3.2.1.15, EC 4.2.2.10)
Source	*Aspergillus niger*
Appearance	Brown liquid
Activity	1,250 PME / g, 150 PL U / g
Units	Pectin-methyl-esterase (PEM)
	Pectin lyase (PL)
Total Protein	22-27 % w/v
Density	1.05-1.20 g per mL
Temperature Range	15–60 °C (operational)
Optimal Temperature	45–60 °C
pH Range	3.5 – 8.0
Optimal pH	5.8 – 6.5
Shelf Life	12–24 months (sealed, recommended storage conditions)

Figure 24

Pectinase technical datasheet

(Source): (*Enzyme Products for Industry, Developers, Researchers and Education*, 2026).

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Parameter	Specification
Enzyme Class	endo-β-1,4-xylosidic bonds (EC 3.2.1.8)
Source	*Aspergillus niger* (non-GM strain)
Appearance	Light brown liquid
Activity	50,000 U / g
Units	Enzyme required to release 1 μmol of xylose from beechwood xylan per min
Density	1.10-1.20 g per mL
Total Protein	10-14 % w/v
Temperature Range	15–55 °C
Optimal Temperature	50–55 °C
pH Range	3.4 – 8.0
pH Optima	4.5 – 5.5
Shelf Life	24 months (sealed, stored when kept recommended storage conditions)

Figure 25

Xylanase technical datasheet

(Source): (*Enzyme Products for Industry, Developers, Researchers and Education*, 2026).

Annex 4. Botanical families

Botanical Family	Common Name	Notable Vegetable Examples
Amaranthaceae	Amaranth family	Spinach, Beetroot, Swiss chard, Quinoa
Amaryllidaceae	Onion family	Onion, Garlic, Leek, Shallot, Chives
Apiaceae	Carrot/Parsley family	Carrot, Celery, Parsnip, Fennel, Cilantro
Asparagaceae	Asparagus family	Asparagus
Asteraceae	Daisy/Sunflower family	Lettuce, Artichoke, Jerusalem artichoke, Endive
Brassicaceae	Mustard/Cole family	Broccoli, Cabbage, Cauliflower, Kale, Radish
Cucurbitaceae	Gourd family	Pumpkin, Zucchini, Cucumber, Melon, Squash
Fabaceae	Legume/Pea family	Peas, Beans, Lentils, Chickpeas, Soybeans
Lamiaceae	Mint family	Basil, Rosemary, Thyme, Oregano, Mint
Malvaceae	Mallow family	Okra
Poaceae	Grass family	Sweet corn, Bamboo shoots
Solanaceae	Nightshade family	Tomato, Potato, Eggplant, Peppers (Bell & Chili)
Convolvulaceae	Morning Glory family	Sweet potato

Figure 26

Botanical families

(Source): based on (Jacobsen, K., 2010) and (Vegetable Families, n.d.).

Annex 5. Vinegar recipes

VEGETABLE LIQUEFICATION

	Ingredient	Quantity	
	Vegetable "x" pulp	1000	g
	Water 1	x	g
	Water 2	200	g
Enzymes	Cellulase	2	g
	Pectinase	2	g
	Amylase	1	g
	Xylanase	1	g

VINEGAR FERMENTATION

Ingredient	Quantity	
Vegetable "x" liquid/juice	1000	g
Alcohol 96%	60	g
Raw unfiltered apple vinegar	180	g

or vinegar from the previous batch as long as it is not pasteurized

PROCEDURE

	Notes:
For the vegetables preparation:	
Peel, wash and/or core	The procedure will work regardless if there is skin or if its raw
Scale weight before cooking	
Apply cooking method and temper	
Scale weight after cooking	The weight difference is "water 1"
For the liquefication method:	All this part can be skept if juicing the vegetables or fruits with a juicing machine
Mix in a thermomix a kilogram of the cool vegetable with the enzymes, water 1 and water 2	
Blitz for about 2 minutes at full speed	
Pour into jars and put in a water bath or oven at 50ºC for 2-4 hours	Letting it sit overnight will improve yield and facilitate filtration
Pass through a thieve/chinois	
For the vinegar:	
If looking for a clearer result pass through a fish-net	
Mix a liter of the passed vegetable liquid with mother vinegar and alcohol	
Pour into a esterilized jar	Fill max. 2/3 of the total volume of the jar
Drop it the air pump stone so it reaches the bottom of the jar	The air pump strenght must be lowered if it bubbles so much that it touches the towel
Seal with a clean towel and a rubber band	
Wait 2 weeks	
Bottling and pasteurization:	
In 1L glass bottles pour 800ml of finished vinegar	Putting too much liquid may make the bottles pop when pasteurizing
Put a lid and drop the bottles in a water bath at 80ºC for 5 minutes	
Instantly pass them into an ice bath	Having a lid on the bottles is what saves all the aromas from being lost
The vinegar is finished and can be stored either in the fridge or at room temperature	

Observations:

The cooking method admits creativity as long as you don't add salt. For example: oils (pumpking, walnut, rapseeds, olives), aromatics (thyme, rosemary, garlic).

The vinegar itself also admits creativity, before pasteurizing put in the bottle (thyme or rosemary, roasted pumpkin seeds, flowers, roasted potato skin, anything that can give it depth).

If you want the flavor of the vinegar to be more intense you can also reduce a part of the base liquid and mix it back in before adding the mother vinegar and alcohol.

Figure 27

Generic vinegar recipe

7. Anexes

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VEGETABLE LIQUEFICATION

	Ingredient	Quantity	
	Celeriac	20	kg
	Water	20%	
Enzymes	Cellulase	0,2%	(2g per kg)
	Pectinase	0,2%	(2g per kg)
	Amylase	0,1%	(1g per kg)
	Xylanase	0,1%	(1g per kg)

VINEGAR FERMENTATION

Ingredient	Quantity	
Roasted celeriac juice	1000	g
Alcohol 96%	60	g
Raw unfiltered apple vinegar	180	g

or vinegar from the previous batch as long as it is not pasteurized

PROCEDURE

For the vegetables preparation:
Roast the celeriac in the oven at 185ºC for 90ºC
Peel and chop into big chunks
Of the entire weight of the cooked celeriac flesh, weight 20% water and the enzymes
In a thermomix put 2kg of celeriac with 400g of water and 12g of the mixed enzymes
Pour the purée into jars, gastronorms, or vacuum bags
Place in a water bath or oven at 50ºC for 2-4 hours
Let sit in the fridge overnight
Pass through a thieve and fish-net

For the vinegar:
Mix the celeriac juice with mother vinegar and alcohol
Pour into a esterilized jar (2/3 full)
Drop in the air pump stone so it reaches the bottom of the jar
Seal with a clean towel and a rubber band
Wait 2 weeks

Bottling and pasteurization:
In 1L glass bottles pour 800ml of finished vinegar
Put a lid and drop the bottles in a water bath at 80ºC for 5 minutes
Instantly pass them into an ice bath
The vinegar is finished and can be stored either in the fridge or at room temperature

Figure 28

Celeriac vinegar recipe

7. Anexes

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VEGETABLE LIQUEFICATION

	Ingredient	Quantity	
	Beetroot	20	kg
	Water	20%	
Enzymes	Cellulase	0,2%	(2g per kg)
	Pectinase	0,2%	(2g per kg)
	Amylase	0,1%	(1g per kg)
	Xylanase	0,1%	(1g per kg)

VINEGAR FERMENTATION

Ingredient	Quantity	
Roasted celeriac juice	1000	g
Alcohol 96%	60	g
Raw unfiltered apple vinegar	180	g

or vinegar from the previous batch as long as it is not pasteurized

PROCEDURE

For the vegetables preparation:

Boil the beetroot until fully cooked
Peel and chop into big chunks
Of the entire weight of the cooked beetroot flesh, weight 20% water and the enzymes
In a thermomix put 2kg of beetroot with 400g of water and 12g of the mixed enzymes
Pour the purée into jars, gastronorms, or vacuum bags
Place in a water bath or oven at 50ºC for 2-4 hours
Let sit in the fridge overnight
Pass through a thieve and fish-net

For the vinegar:

Mix the celeriac juice with mother vinegar and alcohol
Pour into a esterilized jar (2/3 full)
Drop in the air pump stone so it reaches the bottom of the jar
Seal with a clean towel and a rubber band
Wait 2 weeks

Bottling and pasteurization:

In 1L glass bottles pour 800ml of finished vinegar
Put a lid and drop the bottles in a water bath at 80ºC for 5 minutes
Instantly pass them into an ice bath
The vinegar is finished and can be stored either in the fridge or at room temperature

Figure 29

Beetroot vinegar recipe

Annex 6. Beetroot dessert recipe

Beetroot gastric			
Sugar	150 g		Make a caramel with the sugar
Beetroot vinegar	80 g		Deglaze with vinegar and simmer until the caramel has diluted
Apple juice	120 g		Add the apple juice and reduce until glossy
Beetroot vinegar 2	30 g		Infuse with a bouquée of orange peel, chervil and thyme
Orange peel/Chervil/thyme	EQ		Season with some more fresh vinegar
Beetroot gel			
Boiled beetroot juice	500 g		Boil the beetroot until fully cooked, peel and juice
Rhubarb juice	200 g		Mix in the pectin and agar in the beetroot juice at room temperature and bring to a boil for a minute
Apple Juice	300 g		Mix in vigorously the other two juices at room temperature
Sugar	100 g		Pass through a thieve and let set in the molds
Pectin	10 g	0,7%	
Agar	3 g	0,3%	
Beetroot slice			
Beetroot	1000 g		Boil the beetroot until fully cooked, peel and slice at 1mm
Enzymes mix	8 g		Punch out the widest slices at 7cm diameter
Vanilla	1 g		Juice all the trimmings with the enzymes 3 hours at 50ºC and finish 10min at 75ºC
Sugar	120 g		Keep the slices in beets juice with the sugar and vanilla
Roasted walnut compote			
Walnut milk	300 g		Simmer the walnut for 5 min so they remove the tanines of the skin
Roasted walnut	300 g		Sieve and roughtly break the walnut
Brown sugar	100 g		Boil in the milk for 20 min
Salt	EQ		Blitz until smooth and thieve
Roasted walnut foam			
Raw walnut	200 g		Roast 5min at 180ºC and hydrate covered overnight
Water	600 g		Process the nuts with 3 times the amount of water in a Thermomix and pass through a cloth
Dry rice koji	100 g		Add 1 part dry koji for 6 parts milk and incubate at 60ºC for 12h (+10min 75ºC)
Tonka	2 g		Strain in a syphon with a 2% fish gelatin 200 bloom diluted and a charge of NO2
Sugar	90 g		
Glucose DE 84	45 g		

Figure 30

Beetroot dessert written recipe